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Development of Machine Learning Models for Precise Geographic Assignment of Jaguars (*Panthera onca*) Using Complete Genomes and Transfer Learning

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ABSTRACT

Endangered species that most require precise forensic tools for anti-poaching enforcement are typically those for which the fewest genomes are available for model development. In the jaguar (*Panthera onca*), assigning a trafficked individual to its population of origin is constrained both by small, fragmented samples and by a feature-to-sample ratio that makes locus selection challenging. We addressed geographic assignment from SNP data through cross-species transfer learning. We first finetunned DNABERT-2 (**?**) on six felid reference assemblies, and then used the resulting foundation model to compute a label-free Variant Effect Score (VES) for each SNP, a measure of how surprising the alternate allele is in the felid genomic context. These scores initialized per-locus learnable gates in a genotype-matrix multilayer perceptron, which we trained with a differentiable haversine loss to predict coordinates directly. We evaluated every model with 55-fold leave-one-out cross-validation over the jaguar genomes, tuned hyperparameters by 100-trial Optuna Bayesian search, and benchmarked against an identically tuned no-VES control, a fixed-default Locator baseline (**?**), and the SCAT-based assignment of **?**. The Locator baseline reached a median haversine error of 199 km, placing 67.3% of individuals within 500 km, and the best no-VES tuned trial reached 174 km, against a roughly 350–400 km SCAT reference. Introducing VES-initialized gates improved this further: the top-5 trial ensemble reached 165 km median haversine, while the best single Optuna trial reached 149 km and assigned 83.6% of individuals within 500 km of their true origin. Bayesian tuning and cross-species transfer thus contributed successive gains in median error, from 199 to 174 km and from 174 to 149 km. These gains were metric-dependent, however: the VES model lowered the median but not the within-500 km rate, which was highest for the tuned no-VES model (89.1%) rather than for the best single VES trial (83.6%). Taken together, these results show that a foundation model pre-trained across the Felidae can supply constraint signal that a small, fragmented jaguar sample cannot provide on its own.

1 | Introduction

[[TODO]] Intro condensation + tense pass: the Introduction below reuses proposal literature verbatim (jaguar conservation; conservation genomics and ML; the data-scarcity paradox; traditional methods SCAT/SPASIBA/STRUCTURE/KLFDAPC + comparative table + Zenato Lazzari; deep-learning methods Battey-Locator/Degen/Bayliss + comparative table; the transfer-learning promise + foundation models; and the objectives). This is thesis-length. CONDENSE to E&E introduction length — keep the jaguar/forensic motivation, the data-scarcity framing, the transfer-learning rationale, and the objectives; trim the method-by-method literature to citations. Also run a future-tense pass on the flagged passages (the objectives; "The innovative approach proposed here...").

Jaguar Conservation

The jaguar (Panthera onca) is the largest neotropical felid and a keystone apex predator. Historically distributed from the southern United States to Argentine Patagonia the jaguar now occupies only about 46% of its original range, with an estimated 173,000 individuals remaining—roughly half in Brazil (**?**). Jaguars inhabit diverse ecosystems, from dense tropical forests to wetlands, and their protection benefits numerous other species, making them an umbrella species for biodiversity conservation.

Population status varies greatly across biomes. The Amazon and Pantanal harbor the largest, most genetically diverse populations, with extensive connectivity and high habitat suitability. In contrast, the Atlantic Forest and Caatinga populations are small, isolated, and genetically impoverished due to severe fragmentation, while the Cerrado has seen a decline exceeding 50% (**?**). Connectivity corridors, such as along the Araguaia and Paraguay Rivers, are critical to maintain gene flow and avoid local extinctions.

Habitat loss and fragmentation remain the most severe threats. While habitat loss reduces total area, fragmentation more acutely undermines survival by isolating populations and limiting dispersal, especially in smaller groups. In Brazil, most jaguars now occur in protected areas, yet many of these are too small or disconnected to ensure long-term viability. Only in the Amazon do some reserves have sufficient size for multi-century persistence, underscoring the need for connected protected areas and carefully planned dispersal corridors (**?**).

Human-wildlife conflict and illegal trade compound these pressures. Retaliatory killing is common where jaguars prey on livestock, and a growing black market, driven largely by demand from Asian countries for skins, teeth, and bones, targets populations across the Amazon and beyond (**?**). Enforcement is hampered by weak governance, legal loopholes, and clandestine trafficking networks.

A critical gap in conservation is the ability to assign poached individuals to their population of origin, which would allow authorities to identify and prioritize anti-poaching interventions at key hunting hotspots. Strengthening governance, enhancing forensic tools for geographic assignment, expanding protected habitats, and disrupting trafficking routes are therefore essential for the long-term survival of the species.

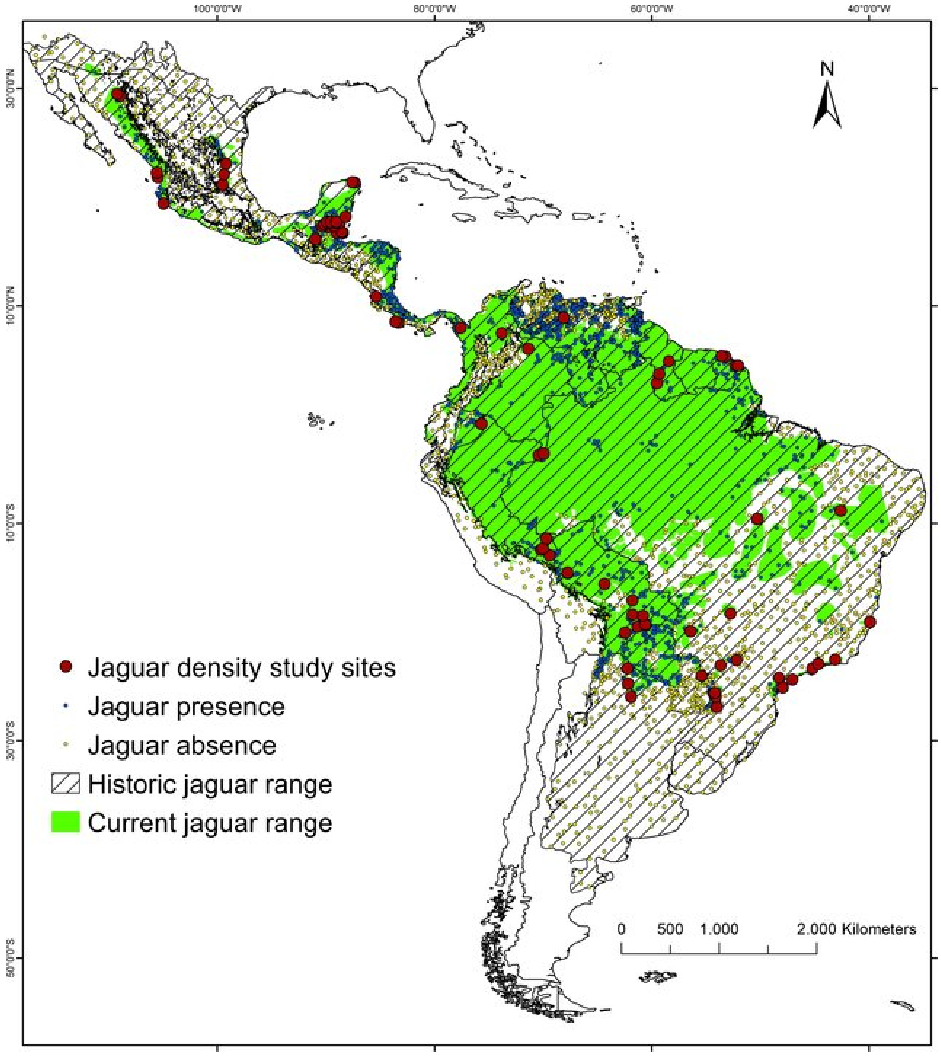

FIGURE 1 | Historical (light shading) and current (dark shading) distribution of the jaguar (*Panthera onca*) across the Americas. Data adapted from (**?**).

Conservation Genomics and Machine Learning

A wide range of disciplines is involved in biodiversity conservation and ecosystem services. Traditionally, genetics has helped elucidate how evolutionary processes shape genotypic and phenotypic variation and determine the population structure of a species, in addition to identifying potential risks related to demographic change and inbreeding on population viability. With the advancement of genetic technology, another new field emerged, genomics, which brought a dramatic increase in the number of genetic markers that should improve the accuracy of diversity estimates and demographic parameters of populations relevant to conservation (**?**).

Among the greatest contributions of genomics to conservation is the ability to predict extinction risks and inform managers about population viability before population decline, allowing decisions on whether translocation is a viable strategy, through the identification of loci and genomic regions responsible for adaptation to local conditions (**?**). Population size is also a key factor in determining demographic viability, and this can be estimated through marker panels designed for individual identification and kinship relationships (**?**). Another area where genomic data is particularly useful and classical genetic techniques are insufficient is the study of past population dynamics of an endangered species – this allows us to elucidate whether the species suffered declines during its past evolutionary history or if its current status is a consequence of what is currently happening.

Machine learning (ML) is increasingly being integrated into conservation genomics to address complex, high-dimensional datasets. ML models can classify individuals into populations of origin, detect cryptic population structures, and predict adaptive traits by learning patterns across thousands or millions of genomic markers.

In addition, ML methods can be used to detect signals of selection, predict future population trends under different environmental scenarios, and integrate genomic, environmental, and movement data for more precise conservation planning. For example, combining genomic data with remote sensing and ecological variables in predictive models can help identify climate-resilient habitats and forecast range shifts. Similarly, unsupervised learning techniques such as clustering and dimensionality reduction can reveal hidden genetic structures that may not be apparent with traditional methods, guiding the design of protected areas and connectivity corridors.

The Challenge of Limited Data in Endangered Species

A critical challenge in conservation genomics is the limited availability of genetic data for endangered species. This scarcity stems from several factors: small population sizes, difficulties in sample collection, ethical considerations, and the high cost of genomic sequencing. Traditional approaches to geographic assignment and population structure analysis often require substantial datasets to achieve reliable results, creating a paradox where the species most in need of conservation attention have the least data available for analysis.

This limitation is particularly acute for jaguars, where population fragmentation and small sample sizes in critical regions (such as the Atlantic Forest) make it difficult to develop robust models for geographic assignment. While SNP-based approaches have provided valuable insights, they often require large numbers of individuals per population to achieve sufficient precision, especially when dealing with subtle genetic differentiation between recently fragmented populations (**??**).

The core challenge addressed by this project is the fundamental paradox in conservation genomics: endangered species, which require the most precise genetic tools for their protection, often have the least genetic data available for analysis. Traditional approaches to geographic assignment require substantial sample sizes per population to achieve reliable results, creating a significant barrier for species like jaguars where populations are small, fragmented, and difficult to sample (**?**).

Traditional Statistical Methods for Geographic Assignment

The scientific challenge of assigning individuals to their geographic origin based on genetic data predates the current wave of machine learning. Classical approaches were built on population genetics theory, statistical modeling, and multivariate analysis. These methods remain foundational in conservation genetics and wildlife forensics, providing the first rigorous frameworks for linking DNA variation to geography. Four representative approaches—SCAT, SPASIBA, STRUCTURE and KLFDAPC—illustrate the logic behind traditional assignment methods.

SCAT – Smoothed and Continuous Assignment Tests The development of SCAT by Wasser and colleagues (2004) (**?**) marked a turning point in applying genetics to wildlife forensics. The core idea of SCAT is to model allele frequencies as continuous surfaces across geographic space rather than discrete population boundaries. By using microsatellite markers from African elephants, SCAT estimated allele frequency gradients through spatial smoothing and then calculated likelihood surfaces for each individual's genotype. This probabilistic framework allowed researchers to infer the most likely geographic origin of ivory samples, even in regions where reference samples were sparse.

The strength of SCAT lies in its ability to interpolate beyond sampled locations, producing continuous assignment maps rather than binary classifications. This was particularly powerful in identifying poaching hotspots in Africa, where many areas lacked dense sampling. Results showed that 50% of samples were assigned within roughly 500 km of their true origin, and resolution improved substantially in well-sampled regions, sometimes below 200 km. However, the method was computationally intensive, requiring Markov Chain Monte Carlo (MCMC) to approximate posterior distributions, and its accuracy decreased markedly in under-sampled regions. Despite these challenges, SCAT established the feasibility of continuous geographic assignment from limited genetic markers and became a cornerstone for forensic applications.

SPASIBA – Spatial Bayesian Inference Building on the ideas of SCAT, Guillot and colleagues (2016) (**?**) developed SPASIBA, a method designed to improve computational efficiency and extend applicability to genome-scale SNP data. SPASIBA treats allele frequencies as spatially autocorrelated random fields, with covariance decaying as geographic distance increases. Instead of relying on MCMC, SPASIBA uses Integrated Nested Laplace Approximation (INLA), a deterministic Bayesian inference method that is far faster and more scalable. This makes it practical to analyze larger datasets with hundreds or thousands of SNPs.

SPASIBA has been applied successfully to a range of taxa, including Florida scrub-jays and *Arabidopsis thaliana*. In the scrub-jay study, using 41 SNPs, SPASIBA localized individuals with a median error of about 26 km, demonstrating impressive resolution at fine scales. In *Arabidopsis*, with 1000 SNPs, the method achieved 75th-percentile errors of around 93 km across the species' European range. These results show that SPASIBA can exploit genome-wide SNP data to generate highly accurate predictions, while also producing probabilistic maps similar to SCAT. Nevertheless, the method assumes Hardy–Weinberg and linkage equilibrium, and like SCAT, its accuracy still depends heavily on reference sampling density. Its computational demands are lower than SCAT's but still non-trivial, requiring specialized software and statistical expertise.

STRUCTURE – Bayesian Clustering for Population Assignment STRUCTURE (**?**) implements a Bayesian clustering framework that infers population structure directly from multilocus genotype data without requiring prior knowledge of individual origins. The model assumes Hardy–Weinberg and linkage equilibrium within populations and uses Markov Chain Monte Carlo (MCMC) sampling to assign individuals probabilistically to K genetic clusters. Each individual receives a vector of membership coefficients, representing the proportion of its genome originating from each cluster, which allows the detection of both discrete population boundaries and admixture.

In practice, STRUCTURE has been widely applied in conservation genetics to identify distinct populations, measure gene flow, and assign individuals of unknown origin to their most likely population. A critical advantage is its ability to reveal hidden structure: clusters can emerge from the genetic data even when population boundaries are not obvious in the field.

Despite its impact, STRUCTURE has limitations. The method produces clusters rather than explicit spatial coordinates, so its geographic resolution is constrained to population-level inferences. The accuracy of assignments depends on the value of K, which must be chosen carefully. Computational cost can also be high for large SNP datasets, though faster variants and related programs such as BAPS and GENELAND have attempted to address this. Moreover, performance declines when populations are weakly differentiated or when reference sampling is sparse.

Nevertheless, STRUCTURE remains a cornerstone in conservation genomics. Its probabilistic framework and ability to detect admixture have made it the default method for many population-level assignment studies. As such, it provides both a conceptual and practical foundation against which newer spatial and machine learning approaches can be compared.

KLFDAPC – Kernel Local Fisher Discriminant Analysis of Principal Components A different branch of approaches focuses on discriminating individuals among predefined populations. One widely used framework is discriminant analysis of principal components (DAPC), which uses PCA for dimensionality reduction followed by linear discriminant analysis to maximize group separation. While DAPC has proven useful, it is inherently linear and may fail to capture nonlinear genetic relationships shaped by geography.

Qin and colleagues (2022) (**?**) addressed this limitation with KLFDAPC, which incorporates kernel methods into the Fisher discriminant framework applied to PCA-transformed data. By embedding data in a nonlinear feature space, KLFDAPC can separate populations that overlap in linear PCA space but differ along nonlinear axes of genetic variation. Applied to both simulations and human SNP datasets, KLFDAPC recovered geographic structures more faithfully than PCA or DAPC, improving classification accuracy and producing axes more consistent with known geography. The method requires labeled reference populations and therefore does not produce continuous spatial assignments like SCAT or SPASIBA. Instead, it excels in cases where discrete populations can be defined and sampled.

Jaguar geographical assignment In 2025, Zenato Lazzari and colleagues (**?**) applied various genomic methods specifically for jaguar conservation, developing a species-specific SNP panel to facilitate geographic assignment. Their study combined whole-genome sequencing with a suite of population genetic tools to create both a full and reduced panel of highly informative SNPs.

The dataset comprised 58 whole genomes representing jaguars from all five Brazilian biomes: Amazon, Pantanal, Atlantic Forest, Cerrado, and Caatinga. From this, they identified candidate SNPs and selected loci based on pairwise F_ST values, maximizing their ability to distinguish between regional populations. The final design produced the JagSNP panel of 459 loci, along with a reduced 84-SNP panel optimized for cost-effectiveness.

To assess performance, the authors used principal component analysis (PCA) to visualize genetic differentiation among individuals from the five biomes and to confirm that the selected SNPs captured population structure. Assignment accuracy was then tested using likelihood-based statistical frameworks. Approximately 98% of individuals were correctly assigned to their biome of origin, and 65–69% could be localized within 500 km of their actual sampling site. Interestingly, the highest accuracy occurred in isolated populations with reduced genetic diversity, while continuous habitats such as the Amazon showed lower resolution due to high gene flow and correlated allele frequencies.

Overall, the Jaguar SNP panel demonstrates how combining genome-wide data, F_ST-based marker selection, PCA validation, and likelihood-based assignment tests can yield a robust and cost-efficient forensic tool for conservation.

Geographic assignment of individuals based on genetic data is an important task in conservation genomics, particularly for forensic applications aimed at combating poaching and illegal trade.

Traditional approaches, such as PCA or Bayesian clustering, have provided valuable insights but often require large sample sizes and struggle with subtle differentiation across fragmented populations.

In recent years, machine learning methods, ranging from kernel-based dimensionality reduction to deep learning, have emerged as powerful alternatives to traditional methods. These approaches improve accuracy, scalability, and the ability to capture non-linear patterns in genomic data.

Deep Learning for Geographical Assignment & Genetic Data Representation

Battey et al – Locator: Deep Neural Network for Geographic Assignment Battey and colleagues (2020) (**?**) set out to address the limitations of traditional Bayesian geographic assignment models, which, while powerful, are computationally expensive and often struggle with fine-scale resolution.

Their goal was to explore whether deep neural networks could offer both greater accuracy and efficiency by directly predicting the geographic coordinates of individuals from their genomic data. By reframing the problem as a regression task, they aimed to capture subtle spatial allele frequency patterns that classical models often miss.

To implement this, the authors designed a fully connected feedforward neural network architecture, essentially a multilayer perceptron (MLP). The input layer took thousands of SNP-derived genotypes encoded as numerical features, which were then passed through several hidden layers with nonlinear activation functions (ReLU) to model complex interactions among alleles. Dropout layers were included to reduce overfitting, and the output layer consisted of two continuous nodes representing latitude and longitude, optimized with Euclidean distance loss between predicted and true coordinates. This relatively simple but powerful architecture allowed Locator to generalize across genomic windows and datasets, balancing predictive capacity with computational efficiency.

To test their approach, the authors applied their model, Locator, to both simulated and empirical data. The simulated datasets provided controlled scenarios where the true spatial origin of each individual was known, allowing for a rigorous assessment of accuracy.

For empirical validation, they used whole-genome sequencing data from a diverse set of systems: Anopheles mosquitoes sampled across Africa, Plasmodium falciparum parasites from Africa and Asia, and global human genomic datasets. This range of datasets was critical, as it tested the method's robustness under varying marker densities, levels of population structure, and geographic scales.

The results demonstrated that the neural network not only outperformed the Bayesian comparator SPASIBA in terms of

TABLE 1 | Comparison of traditional methods for geographic assignment

Method	Input Data	Output Type	Comp. Cost	Strengths	Weaknesses
SCAT	Geo-tagged genetic data, continuous allele frequencies	Continuous origin estimates	Moderate-high	Good for fine spatial gradients, continuous mapping	Sensitive to sampling density, assumptions about dispersal
SPASIBA	SNP / genotype data + spatial coordinates	Continuous geographic assignment	High	Provides high resolution continuous estimates	Computationally intensive, requires many samples
STRUCTURE	Multilocus genotype data, no spatial info built in	Assigns individuals to discrete clusters	Moderate	Captures population structure, widely used	Coarse, less precise for continuous geography
KLFDAPC	Genotypes + optional geography, dimension reduction & clustering	Discrete or semi-continuous assignment	Moderate	Can combine spatial smoothing & allele frequency variation	Less interpretable, may miss fine-scale continuous gradient

speed but also achieved strikingly higher accuracy. Locator localized individuals to within approximately four dispersal generations in simulated data, while in empirical datasets median assignment errors ranged from only 5.7 km in mosquitoes to about 85 km in Plasmodium and human populations. Importantly, the runtime improvements were substantial, with Locator operating an order of magnitude faster than SPASIBA, making it suitable for genome-wide applications.

Despite these advances, the study highlighted two key limitations: the model's dependence on large, representative training datasets and its black-box nature, which obscures the biological interpretation of the features driving predictions.

Degen et al – Comparing Machine Learning Approaches for Continuous Spatial Assignment Degen and colleagues (2025) (**?**) investigated whether modern machine learning approaches could improve the continuous geographic assignment of individuals from SNP data in European tree species. Their goal was to benchmark several methods against each other, including deep learning, Gaussian process regression (GPR), k-nearest neighbors, and genomic best linear unbiased prediction (GBLUP).

The deep learning component was built around a multilayer perceptron (MLP) similar in spirit to Battey et al.'s Locator, but optimized for continuous regression of geographic coordinates. Genotype features (SNPs) were passed into a network of dense hidden layers with nonlinear activations, designed to capture allele frequency gradients and complex dependencies between loci. Hyperparameters such as layer depth, number of units per layer, and dropout rates were tuned to balance generalization with predictive performance. The output layer produced two continuous values for latitude and longitude, again trained with a regression loss function that minimized the Euclidean distance between predicted and true coordinates. This allowed the DNN to flexibly model nonlinear genotype–geography relationships, while complementing GPR, which excelled at uncertainty estimation.

The datasets used were SNP panels from two widespread species: European beech (*Fagus sylvatica*) and European oak (*Quercus robur*). Both species were sampled broadly across their ranges, providing thousands of individuals with known geographic origins. These large datasets offered an opportunity to test whether machine learning could recover fine-scale population structure and spatial gradients of allele frequencies that reflect evolutionary and demographic processes.

Results revealed that both deep neural networks (multilayer perceptrons) and GPR with a Matérn kernel achieved the highest accuracy, predicting origins with median errors of ~55–76 km for beech and ~263–278 km for oak. These errors were substantially smaller than those achieved by simpler models like k-NN or GBLUP.

The authors noted that GPR provided probabilistic uncertainty estimates, which are useful in practice, while DNNs were more flexible in modeling nonlinear interactions between SNPs. Both methods required large amounts of training data and computational resources.

Bayliss et al. - Hierarchical ML for Geographic Assignment of Genomic Samples A recent preprint by Bayliss et al. (2023) (**?**) introduces a hierarchical machine learning framework for predicting the geographic origin of genomic samples, with Salmonella enterica serovar Enteritidis as the proof of concept. The aim is to trace sources of infection by assigning each genome to its most likely continent, sub-region, and country of origin directly from DNA data. Rather than relying on deep neural networks, the authors explored more classical ML approaches such as random forests, gradient boosting, and support vector machines, demonstrating the strength of these methods in large-scale genomic inference.

The dataset consisted of 2,313 S. Enteritidis genomes collected by the UK Health Security Agency between 2014 and 2019, each linked to one of 53 geographic classes organized hierarchically: four continents, eleven sub-regions, and thirty-eight countries. Instead of training a single flat classifier, the researchers implemented a "local classifier per node" strategy, cascading models so that predictions flowed from continent to sub-region to country. This design reduced misclassifications by ensuring that finer-level predictions were restricted to plausible sub-areas once higher-level categories were determined.

Genetic features were extracted from sequencing data using k-mer units, and multiple algorithms were systematically evaluated under consistent conditions. The team tested random forests, SVMs, k-nearest neighbors, XGBoost, and Naïve

Bayes, while carefully addressing class imbalance with resampling techniques and reducing feature dimensionality through iterative selection based on random forest importance scores. Ultimately, optimized random forests were selected at each hierarchical level, striking a balance between predictive accuracy, interpretability, and computational speed. The resulting pipeline could analyze raw reads and deliver a geographic prediction in under four minutes—a stark contrast with more labor-intensive phylogeographic analyses.

The results were compelling. At the continental level, accuracy was extremely high (macro F1 $\approx$ 0.954). Performance remained strong at the sub-regional level (F1 $\approx$ 0.718) and reached a respectable macro F1 of 0.661 at the country level, even across 38 classes. In practice, some countries—particularly those with high travel links to the UK—achieved hierarchical F1 scores above 0.90, reflecting near-perfect precision. Crucially, the method generalized well: when applied to external Salmonella datasets outside the original training set, accuracy remained robust, indicating that the models captured true geographic signals in the genomes rather than simply memorizing the data.

The novelty of this work lies in showing that established ensemble methods, implemented hierarchically, can deliver accurate, fine-scale geographic predictions from genomic data without the heavy computational demands of deep learning. Bayliss et al.'s approach enriches the methodological toolkit for geographic assignment, complementing deep-learning tools like Locator and Gaussian-process models with a practical, transparent, and efficient alternative.

The Promise of Transfer Learning in Conservation Genomics

Recent advances in machine learning, particularly in the field of natural language processing, have introduced powerful techniques for learning from limited data through transfer learning. These approaches allow models pre-trained on large, general datasets to be fine-tuned for specific tasks with much smaller datasets, often achieving superior performance compared to models trained from scratch.

In the context of genomics, foundation models like DNABERT-2 (**?**) have demonstrated remarkable capabilities in understanding DNA sequence patterns across multiple species. These models can capture complex genomic features and relationships that may not be apparent through traditional analytical methods. By leveraging the knowledge learned from large, diverse genomic datasets, these models can potentially overcome the data scarcity challenges faced in endangered species conservation.

Recent developments in genomic machine learning have introduced foundation models, which are large-scale pretrained architectures designed to capture universal features of DNA sequences across species. These models—such as **DNABERT-2** (**?**) and the **Nucleotide Transformer** (Dalla-Torre et al., 2025) (**?**)—represent a paradigm shift in the field. Their goal is to learn rich, generalizable representations of nucleotide sequences that can be transferred to diverse downstream tasks, from variant effect prediction to motif discovery, often with minimal fine-tuning data.

The results have been striking. Both DNABERT-2 and the Nucleotide Transformer achieve state-of-the-art performance across a wide range of genomic benchmarks. DNABERT-2, for instance, outperforms classical CNNs and RNNs on promoter prediction, transcription factor binding site recognition, and splice site identification, often by large margins (**?**). Similarly, the Nucleotide Transformer has shown that pretraining across many species enables cross-species transfer, where a model trained on non-human genomes can still perform strongly on human tasks (**?**).

The innovative approach proposed here leverages the power of transfer learning in machine learning to overcome this data scarcity challenge. By pre-training a foundation model on comprehensive feline genomic data and then fine-tuning it on jaguar-specific sequences, we can potentially achieve levels of precision and reliability that were previously unattainable with traditional methods. This approach represents a paradigm shift in conservation genomics, moving from methods that require large datasets to methods that can work effectively with limited data through the power of pre-trained models.

OBJECTIVES

General Objective Develop and evaluate a machine learning pipeline using transfer learning and foundation models for precise geographic assignment of jaguars (*Panthera onca*) from different Brazilian biomes, overcoming the limitations of limited genetic data through pre-training on comprehensive feline genomic datasets.

Specific Objectives

1. Pre-train a DNABERT-2 foundation model on comprehensive feline genomic data from the Darwin's Ark database to learn general patterns of genomic variation across felid species;

2. Fine-tune the pre-trained model on complete jaguar genomes from different Brazilian biomes to develop a specialized model for jaguar geographic assignment;

3. Evaluate the performance of the transfer learning approach compared to traditional methods for geographic assignment, particularly in scenarios with limited sample sizes;

4. Develop a computational pipeline for automated geographic assignment of jaguar samples of unknown origin, with applications in forensic analysis and conservation management;

5. Assess the model's ability to identify subtle genetic differentiation between recently fragmented populations, particularly in the Atlantic Forest biome where sample sizes are critically small.

2 | Materials and Methods

2.1 | Foundation Model: DNABERT-2

We will employ the DNABERT-2 architecture as our foundation model, which represents the current state-of-the-art in genomic language modeling (**?**).

The model incorporates several advanced architectural features designed to handle the specific complexities of genomic data. Central to its design is Attention with Linear Biases (ALiBi), a mechanism that allows the model to extrapolate to longer input sequences than those seen during training,

TABLE 2 | Comparison of machine learning approaches for geographic assignment

Study	Goals	Methods	Results	Limitations
Battey et al. (2020)	Improve speed and accuracy of geographic assignment	Deep neural network (Locator) predicting coordinates	Median errors 5.7–85 km; 10× faster than Bayesian models	Requires large datasets; limited interpretability
Degen et al. (2025)	Benchmark ML models for continuous assignment	DNNs, GPR, k-NN, GBLUP	DNN and GPR: 55–76 km (beech), 263–278 km (oak)	High computational demands; requires large datasets
Bayliss et al. (2022)	Hierarchical geographic assignment	Random Forests, SVMs, XGBoost (hierarchical)	F1: 0.954 (continent), 0.718 (sub-region), 0.661 (country)	Performance drops at finer scales; requires substantial data

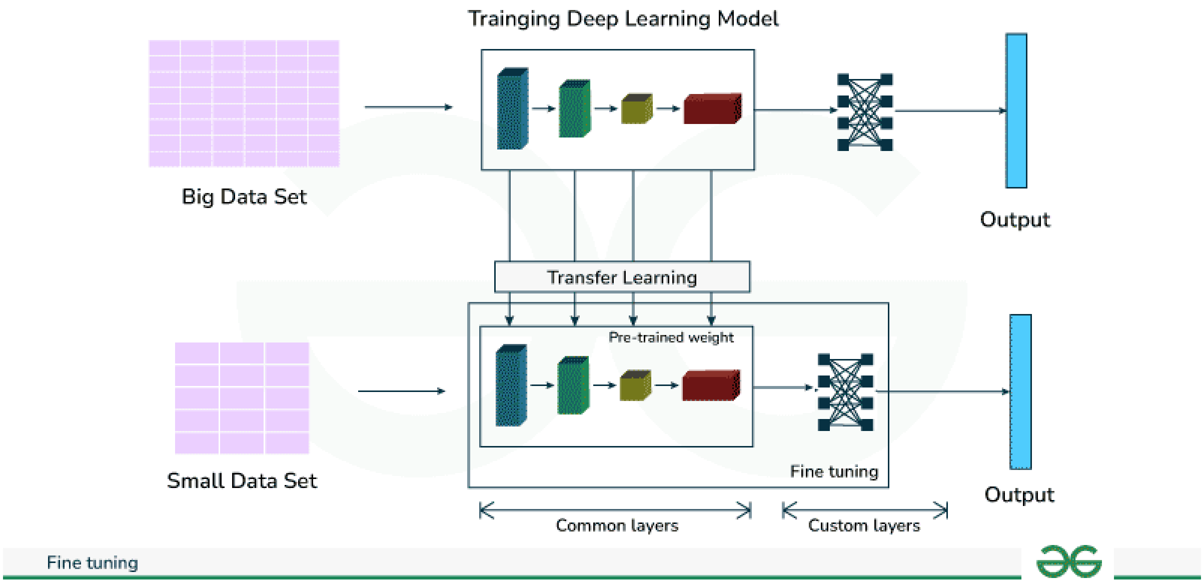

FIGURE 2 | Illustration of transfer learning and fine-tuning : a pre-trained base model's shared layers are reused, while task-specific layers are added and trained on the target dataset. Understanding Fine-Tuning of Large Language Models (LLMs) : Instruction Tuning and Beyond (Medium, 2023).

effectively bypassing the limitations of traditional learned positional embeddings. Furthermore, the integration of Flash Attention drastically improves memory usage and computational speed during both training and inference phases. These optimizations, combined with its inherent multi-species pre-training capability, allow DNABERT-2 to learn robust, generalizable patterns across diverse genomic datasets before specific adaptation.

2.2 | Felid Foundation Pre-training Corpus

We assembled the felid foundation pre-training corpus from six reference genome assemblies spanning three genera of the family Felidae, chosen so that the jaguar (*Panthera onca*) sits within a phylogenetically broad but taxonomically focused sample of its own family rather than within a generic vertebrate mixture. The roster is closed and pinned in code: each entry fixes the species binomial, an assembly identifier, the assembly name, and the NCBI taxonomy identifier, and adding a species requires an explicit code and test change, so that the corpus cannot silently drift between runs. Five of the six assemblies were drawn from NCBI RefSeq and downloaded from the canonical RefSeq FTP tree; the jaguar assembly was taken from the DNA Zoo Hi-C submission (`Panthera_onca_HiC`) through a pinned direct URL, with a HuggingFace mirror as a secondary source. Every compressed FASTA was integrity-checked at download time against a manually pinned checksum — MD5 for the RefSeq files and SHA-256 for the jaguar DNA Zoo file — so that a truncated or stale payload is rejected rather than silently ingested. Table **??** lists the six assemblies together with the number of 512 bp windows each contributed to the corpus.

The corpus was built directly from the reference FASTA files, with no variant-calling or consensus step: we emitted one sequence record per contig, tagged with a provenance label of `reference` and a species slug (for example `panthera_onca`) that identifies its source assembly. Each contig sequence was first normalised by upper-casing and validating it against the fixed nucleotide alphabet $\{A, C, G, T, N\}$; because the foundation configuration pins the unsupported-symbol policy to `reject`, any character outside this alphabet aborts the run rather than being coerced, so that no out-of-alphabet base can leak into a training window. Contigs shorter than the 512 bp context window were dropped as short sequences before windowing; together with the ambiguity filter, this reduced the jaguar Hi-C assembly from 410,593 contigs to

TABLE 3 | The six felid reference assemblies in the foundation pre-training corpus. Windows are the number of tokenised 512 bp windows contributed to the train and validation folds; species are listed in the identifier-sorted registry order used by the pipeline.

Species	Common name	Identifier	Assembly	Train windows	Validation windows
Panthera onca	Jaguar	`DNAZOO_Panthera_onca_HiC`	`Panthera_onca_HiC`	4,822,435	1,217,303
Felis catus	Domestic cat	`GCF_000181335.3`	`Felis_catus_9.0`	5,014,368	1,264,395
Panthera tigris	Amur tiger	`GCF_000464555.1`	`PanTig1.0`	4,384,605	1,105,029
Panthera pardus	Leopard	`GCF_001857705.1`	`PanPar1.0`	4,240,897	1,080,228
Puma concolor	Puma	`GCF_003327715.1`	`PumCon1.0`	3,763,602	935,831
Panthera leo	Lion	`GCF_018350215.1`	`P.leo_Ple1_pat1.1`	4,739,028	1,188,593
			Total	26,964,935	6,791,379

67,420 windowing-eligible sequences.

We then slid a fixed 512 bp window along each retained contig with a stride of 384 bp — that is, a 128 bp overlap between consecutive windows — which provides coverage of window-boundary regions while bounding the total corpus size. Windows were never N-padded at contig ends: any trailing fragment shorter than 512 bp was discarded rather than padded, so that every emitted window is exactly 512 bp of real sequence. As a further quality filter we required each window to contain at most a 5% fraction of ambiguous (N) bases, discarding those that exceeded this threshold. This procedure produced 33,756,314 tokenised windows in total across the six assemblies.

Each retained window was tokenised with the DNABERT-2 byte-pair-encoding (BPE) tokenizer (**?**), loaded from the HuggingFace model `zhihan1996/DNABERT-2-117M` and pinned to the immutable Git revision `7bce263b15377fc15361f52c fab88f8b586abda0` so that the tokenisation is exactly reproducible; as DNABERT-2's custom implementation requires, the tokenizer was loaded with `trust_remote_code` enabled. Tokenisation added the model's special tokens and, rather than truncating, asserted that no window exceeded the model's maximum position-embedding length of 512 tokens, so that any window whose BPE encoding overflowed the context would fail loudly instead of being silently clipped. The tokenised windows were written to a partitioned Parquet corpus keyed by (split, contig, block), each window carrying its genomic coordinates and a SHA-256 hash of its nucleotide string, so that any window can later be audited against its source assembly without re-reading the original FASTA. To keep peak memory bounded on the six multi-gigabase assemblies, each species was processed independently and streamed to the writer in fixed-size chunks, and a cross-species contig-name collision check guaranteed that the shared `contig:block_start-block_end` locus namespace never aliases windows from different assemblies.

Train and validation folds were assigned in a locus-safe manner that eliminates leakage between overlapping windows. Before windowing, each assembly's coordinate space was partitioned into contiguous 50 kb locus blocks, and every block was assigned deterministically to a fold by hashing its locus identifier: we computed a SHA-256 digest of the string `feline-locus-split-v1:{locus_id}`, mapped its leading 64 bits into $[0, 1)$, and applied an 80%/20% cumulative-weight cutoff to route the block to either train or validation. Because every window inherits the split of the single 50 kb block that contains it, and the 50 kb block size comfortably exceeds the 512 bp window length, no two overlapping windows can ever straddle the train/validation boundary; the pipeline additionally asserts this invariant explicitly, raising a hard error if any locus maps to two folds or if overlapping windows on the same contig receive different folds. The 50 kb block also exceeds the autocorrelation length of most local genomic features, so that the validation fold measures generalisation to genuinely unseen loci rather than to shifted copies of training windows. The resulting global split placed 26,964,935 windows (79.9%) in the training fold and 6,791,379 (20.1%) in the validation fold.

We then adapted DNABERT-2 to the felid corpus by continued masked-language-modelling (MLM) pre-training, warm-starting from the same pinned `zhihan1996/DNABERT-2-117M` checkpoint and revision and loading it through `AutoModelForMaskedLM` so that the MLM prediction head was available. Following the standard MLM objective, 15% of tokens were masked dynamically in every batch by a data collator — a fresh random mask at each step — and the model was trained to recover the original tokens from their surrounding context. Training used bfloat16 (bf16) mixed precision under `accelerate` together with the AdamW optimizer (**?**), with decoupled weight decay of 0.01 applied to all parameters except biases and LayerNorm weights, an initial learning rate of 5×10^{-5} — a conservative rate appropriate for continued rather than from-scratch pre-training — and a cosine-annealing schedule preceded by a 500-step linear warmup. Because DNABERT-2 ships without a padding token, we set the pad token to the existing end-of-sequence token. DNABERT-2's Triton flash-attention kernel was disabled at load time, as it is incompatible with the installed Triton version, and the model fell back to the standard PyTorch attention path. The remaining optimisation settings are collected in Table **??**; the effective batch size is $32 \times 2 \times W$ sequences, where W is the number of data-parallel processes. Under this schedule the run reached its best validation MLM cross-entropy loss of 4.278 at step 36,000 — a validation perplexity of approximately 72 — with the most recent recorded checkpoint at step 40,000 and an early-stopping patience counter of four out of five, that is, four consecutive 1,000-step evaluation cycles without improvement over the step-36,000 best.

TABLE 4 | Continued masked-language-modelling hyperparameters for felid foundation pre-training.

Hyperparameter	Value
Warm-start checkpoint	`zhihan1996/DNABERT-2-117M` @ `7bce263b...abda0`
Objective	dynamic MLM, mask probability 0.15
Maximum sequence length	512 tokens
Optimizer	AdamW (weight decay 0.01; excl. bias, LayerNorm)
Learning rate	5×10^{-5}
LR schedule	cosine annealing, 500-step linear warmup
Maximum training steps	50,000
Per-device train batch size	32
Gradient accumulation steps	2
Per-device eval batch size	32
Mixed precision	bf16
Gradient clipping (max L2 norm)	1.0
Evaluation frequency	every 1,000 steps
Maximum eval steps per cycle	500
Checkpoint frequency	every 1,000 steps
Logging frequency	every 10 steps
Early-stopping patience	5 evaluation cycles
Pad-token fallback	end-of-sequence token
Random seed	42
DataLoader workers	8
Shuffle-buffer size	8,192

2.3 | Jaguar Samples and Study Populations

Our geographic-assignment experiments drew on the whole-genome jaguar (*Panthera onca*) resequencing panel assembled by **?**. Rather than re-derive genotype calls from the raw reads, we started from the authors' curated, multi-sample variant call file (VCF), `jaguar.57samples.allChr.snps.hardFilter.bi.maf.ld.masked.hwe.recode.vcf` (~147 MB; 147,354,171 bytes). As its filename tokens record, this file holds biallelic single-nucleotide polymorphisms (SNPs) across all chromosomes, retained after hard filtering and after pruning for minor-allele frequency, linkage disequilibrium, masking, and Hardy–Weinberg equilibrium, and it reports 57 sequenced individuals. The dataset was originally deposited on DataDryad (doi:10.5061/dryad.4tmpg4fkm) under a CC0 license; for reproducibility we retrieved a byte-identical mirror from the public HuggingFace repository `coding-racoon/jaguar-raw-data` and verified it against a pinned SHA-256 digest (`8bf9083a...93308b`) before use. Each individual's geographic origin was taken from a companion metadata table, `jaguar_location.csv`, mirrored on the same repository, which provides for every sample a decimal-degree latitude and longitude (WGS-84 datum) together with a categorical biome-of-origin label. Our loader required the columns `sample_id`, `individual_id`, `biome_population_label`, `latitude`, and `longitude`.

We joined the samples listed in the VCF header to the metadata table on `sample_id`, keeping only individuals present in both—an inner join. Two of the 57 sequenced samples, `LegadoSP` and `bPon529`, carried no matching metadata row and were therefore excluded, leaving 55 individuals as the modelled set. Every retained sample mapped to a distinct `individual_id`, so the study comprised 55 unique jaguars, each contributing a single genome. We note that the deposited VCF records 57 sequenced individuals, whereas **?** describe a panel of 58 whole genomes; we did not attempt to resolve this one-genome discrepancy beyond noting it, and worked throughout from the 55 individuals that remained after the metadata join. From these individuals we assembled a dense genotype matrix over the 83,527 biallelic SNP loci that survived the in-pipeline VCF checks—a FILTER of `PASS` or `.`, single-nucleotide REF and ALT alleles drawn from $\{A, C, G, T, N\}$, and no multi-allelic sites—yielding a $55 \times 83{,}527$ matrix of alternate-allele dosages (0, 1, or 2 per diploid genotype).

The 55 individuals were distributed across the five biomes that define our study populations. These labels formed a closed, alphabetically ordered vocabulary—Amazon, Atlantic Forest, Caatinga, Cerrado, and Pantanal—with no "unknown" catch-all class: any metadata row bearing a label outside this set was rejected outright. Table **??** reports the per-biome sample sizes for the retained individuals. The distribution was markedly uneven, ranging from 5 individuals in the Caatinga to 18 in the Amazon, and the two smallest populations (Caatinga, $n = 5$; Pantanal, $n = 6$) together accounted for only 11 of the 55 genomes. The sampled localities spanned roughly 25.6°S to 2.5°N in latitude and 70.2°W to 40.1°W in longitude. Taken together, these features—few individuals per population, an order-of-magnitude imbalance across biomes, and tens of thousands of candidate loci—define the small-sample, high-dimensional regime that motivated the transfer-learning and locus-selection strategies described in the following subsections.

2.4 | Variant Effect Scoring

Having pre-trained the felid foundation model, we repurposed it as a frozen scorer that assigns each jaguar SNP a label-free measure of functional importance, which we term the Variant Effect Score (VES). The intuition is straightforward. A

TABLE 5 | Per-biome sample sizes for the 55 modelled jaguar genomes, derived from the biome labels in the joined metadata. Counts sum to the full modelled set.

Biome	Individuals (n)
Amazon	18
Atlantic Forest	13
Caatinga	5
Cerrado	13
Pantanal	6
Total	55

genomic language model trained by masked language modelling learns, for every position, a conditional distribution over nucleotides given the surrounding context; a variant whose alternate allele is assigned low probability under that distribution sits in a region the model regards as evolutionarily constrained across the Felidae, which we hypothesised makes it a stronger candidate to carry biome-discriminating signal than a variant in a region the model treats as freely substitutable. Crucially, because this constraint estimate is derived from cross-species reference genomes rather than from the small jaguar sample, it does not degrade as the number of individuals per biome shrinks — precisely the regime in which frequency-based locus selection becomes unreliable.

We stress that this is a working hypothesis rather than an established fact. Population-genetic theory offers a competing expectation: purifying selection tends to reduce within-species, between-population differentiation at constrained sites, which would make such loci *less* informative for geographic assignment rather than more. Our results do not adjudicate between these mechanisms — in particular, we did not verify that high-|VES| loci come to dominate the trained gates — so VES is best read as an empirically motivated prior on locus importance whose biological interpretation remains open.

Window construction. For each biallelic SNP we reconstructed the local genomic context using the same window-extraction primitive as the fine-tuning corpus. Given the 1-based VCF position p, we took the 0-based locus index $\ell = p - 1$ and cut the half-open interval $[\ell - 256,\ \ell + 256)$ from the reference contig, that is, 256 upstream bases, the centre base, and 255 downstream bases, for a total of 512 characters. The reference allele was written at the centre position and the entire window was upper-cased to the `ACGTN` alphabet. Windows that would have extended past either contig boundary were *not* N-padded; instead the locus was skipped, so that no synthetic flank could contaminate a score. All windows were extracted against the DNA Zoo *Panthera onca* HiC assembly (`data/raw/felid_foundation/reference/DNAZOO_Panthera_onca_HiC.fna.gz`), the same assembly configured throughout the fine-tuning pipeline.

Tokenization and centre masking. Each window was tokenized with the DNABERT-2 byte-pair-encoding tokenizer (**?**) bundled with the felid checkpoint, with offset mapping enabled (`return_offsets_mapping=True`), padded to the batch and truncated at a maximum length of 512 tokens. Because byte-pair encoding merges variable-length nucleotide runs into single tokens, the variant does not occupy a fixed token index; we therefore located it through the offset mapping, selecting the token whose character span [start, end) contains the centre character index 256 and skipping the special tokens (`CLS`, `SEP`, `PAD`), which carry the zero-width span $(0, 0)$. The identified centre token was then replaced by the tokenizer's mask token, recreating the masked-prediction task the model was trained on: the model sees the flanking context but neither the reference nor the alternate base at the variant position.

Score definition. We loaded the felid-pretrained checkpoint (`models/foundation_felid/best/hf_model`) with the masked-language-model head via `AutoModelForMaskedLM` rather than the bare encoder, so that the vocabulary-projection logits are exposed. All parameters were frozen (`requires_grad=False`), the model was placed in evaluation mode, and every forward pass was run under inference mode. For a masked window we took the model's logits at the centre position, converted them to log-probabilities with a log-softmax over the vocabulary (computed in float32), and defined the score as the log-likelihood ratio of the alternate against the reference allele,

$$\mathrm{VES}(\ell) \;=\; \log P\big(x_\ell = a_{\mathrm{alt}} \mid x_{\setminus\ell}\big) \;-\; \log P\big(x_\ell = a_{\mathrm{ref}} \mid x_{\setminus\ell}\big)\,, \tag{1}$$

where x_ℓ is the masked centre token, $x_{\setminus\ell}$ the surrounding context, and a_{ref}, a_{alt} the reference and alternate alleles. Each single-base allele was mapped to a vocabulary index by encoding it in isolation (`add_special_tokens=False`) and taking the first resulting token id, and the two indices were then read off the same masked-position log-probability vector. The sign of the score is interpretable directly: a negative VES indicates that the alternate allele is unexpected in the felid genomic context (a constrained position), a value near zero indicates that the two alleles are roughly interchangeable (an unconstrained position), and a positive value indicates that the alternate is more expected than the reference base carried by the assembly.

Coverage and score distribution. We scored the full panel of 83,527 biallelic loci from the genotype matrix in batches of 128 windows. A locus received VES = 0.0 only when a score could not be computed — a missing contig, a boundary-truncated window, an unlocatable centre token, or an allele that failed to encode — and in practice this affected a single locus, so that 83,526 of 83,527 loci (99.999%) carried a model-derived score. Across the panel the scores were approximately centred at zero with a heavy negative tail: mean −0.120, population standard deviation 2.544, minimum −20.495, and maximum 15.690. This long negative tail is the signal of interest, corresponding to the loci where the alternate allele is most strongly disfavoured by the felid model.

Caching. Because scoring the panel requires a full forward pass of the frozen transformer per locus, and its result is independent of any downstream training choice, we computed the scores once and cached them. The per-locus score tensor was written to `ves_scores.pt` and its summary metadata — locus count, scored and failed counts, and the distributional statistics reported above — to `ves_scores.json`, both under the shared genotype-matrix cache directory (`data/processed/genotype_matrix`). All subsequent runs detected the cached tensor and reloaded it rather than re-invoking the backbone, so that every VES-integration strategy evaluated downstream operated on an identical, reproducible set of scores.

2.5 | Genotype Matrix Construction

We represented each jaguar as a vector of allele counts over the biallelic single-nucleotide polymorphism (SNP) loci that survived filtering, and stacked these vectors into a dense genotype matrix of shape (individuals × loci). Whereas the DNABERT-2 embedding pathway summarises sequence context, this representation deliberately preserves every allelic state — homozygous reference calls included — because for population-level geographic assignment the absence of a variant is as diagnostic as its presence. We built the matrix directly from the multi-sample VCF (`jaguar.57samples.allChr.snps.hardFilter.bi.maf.ld.masked.hwe.recode.vcf`) in a single streaming pass and cached it to disk as an `Int8` tensor alongside per-locus and per-individual metadata, so that it was computed once and reused unchanged across every experimental run.

Allele-count encoding. We encoded each diploid genotype as the sum of its two allele indices, so that the reference allele contributes 0 and the single alternate allele contributes 1. This yields the standard 0/1/2 alternate-allele-count scheme summarised in Table **??**: homozygous-reference calls (0/0) map to 0 and are retained rather than discarded, heterozygous calls (0/1 or 1/0) map to 1, and homozygous-alternate calls (1/1) map to 2. Missing calls were set aside with the sentinel value −1 prior to imputation. When splitting the `GT` field we accepted both unphased (`/`) and phased (`|`) genotype separators.

Locus filtering. A VCF record was admitted as a column of the matrix only if it passed every one of the following criteria. First, its `FILTER` field had to be either `PASS` or the missing-value marker `.`. Second, the site had to be strictly biallelic: we skipped records whose `ALT` field contained a comma, and, after collapsing the VCF monomorphic sentinel (a lone `.` in `ALT`, which normalises to zero alternate alleles), required exactly one alternate allele. Third, both `REF` and `ALT` had to be a single character drawn from the allowed nucleotide alphabet {A, C, G, T, N} (case-insensitive). Together, this single-nucleotide requirement and the biallelic constraint implicitly exclude indels, multi-allelic sites, spanning deletions (encoded as *), and monomorphic records, without any dedicated special-casing. Finally, the `FORMAT` field had to declare a `GT` key, from which we read the per-sample genotype.

Sample selection. We retained only samples present in both the VCF `#CHROM` header and the metadata CSV, joining the two on sample identifier. This inner join produced 55 matched individuals, each carrying its individual identifier, biome population label, and decimal (WGS-84) latitude and longitude. The two VCF samples absent from the metadata (`LegadoSP` and `bPon529`) were dropped with a logged warning, reducing the 57 sequenced samples to the 55 used downstream.

Missing-data handling. We treated a genotype as missing (encoded −1) whenever the `GT` field was empty, contained any `.` no-call allele token, or was absent from the sample's colon-delimited values. Genotypes that were syntactically valid but not diploid — that is, with a token count other than two — were likewise demoted to missing rather than encoded. Non-numeric, non-missing allele tokens were instead treated as a hard error, so that a corrupted VCF failed loudly rather than being silently coerced.

Per-fold allele-frequency imputation. Because leave-one-out cross-validation (LOOCV) evaluates one held-out individual at a time, we imputed missing values independently inside each of the 55 folds using only that fold's training individuals, which prevents information about the held-out sample from leaking into the imputation. For each locus we estimated the alternate-allele frequency from the non-missing training calls as the sum of their encoded values divided by twice the number of non-missing training individuals; a locus with no non-missing training observation defaulted to a frequency of 0 and was imputed as homozygous reference. We then filled each missing entry — for training and held-out individuals alike — by summing two independent Bernoulli draws at that frequency, producing an imputed value in $\{0, 1, 2\}$. This binomial draw at the training allele frequency follows the imputation procedure of Locator (**?**). To keep imputation reproducible, we seeded a dedicated random generator per fold as the base seed plus the fold index (base seed 42).

Final matrix. The construction produced a dense `Int8` genotype matrix of 55 individuals × 83,527 biallelic SNP loci. This matrix is the shared input from which all downstream Variant Effect Scoring integration modes and MLP training runs were derived. We note that the locus panel itself — the 83,527 SNPs, together with the upstream minor-allele-frequency, linkage-disequilibrium, masking and Hardy–Weinberg pruning already applied when the source VCF was assembled — was defined once over all 55 individuals rather than re-derived inside each leave-one-out fold. This filtering is label-free (it does not use the geographic coordinates), so it does not leak the held-out label; it does, however, mean the candidate-locus set is common to every fold.

2.6 | Geographic Assignment Model

The geographic assignment task will be formulated as a regression problem, where the model predicts the latitude and

Genotype	Encoding	Meaning
`0/0`	0	Homozygous reference (retained)
`0/1`	1	Heterozygous
`1/1`	2	Homozygous alternate
`./.`	−1	Missing (imputed per fold)

TABLE 6 | Allele-count encoding of diploid genotypes. The encoded value is the sum of the two allele indices; missing entries carry the sentinel −1 until per-fold imputation.

longitude of origin for each jaguar genome, similar to the approach used by *Locator* (**?**). This allows for finer spatial resolution than biome classification and enables targeted enforcement at critical poaching hotspots.

The geographic assignment stage mapped each individual's genotype vector directly onto a geographic coordinate, using a compact multilayer perceptron (MLP) in the spirit of Locator (**?**) and GeoGenIE

[[TODO]] EXTEND: add reference for GeoGenIE (Martin et al., 2025)

. The network took as input a per-individual vector of allele counts, one entry per retained single-nucleotide polymorphism (SNP) locus, with values in $\{0, 1, 2\}$ for the homozygous-reference, heterozygous, and homozygous-alternate states; once variant-effect-score (VES) weighting was applied, these entries became continuous. In the full-locus configuration behind the reported model, this vector spanned all 83,527 loci that survived upstream filtering, evaluated across the 55 jaguar individuals.

When VES integration was set to *learnable* mode, the genotype vector first passed through a per-locus sigmoid gate before any other operation, $x_\ell = g_\ell \cdot \sigma(w_\ell)$, where g_ℓ is the allele count at locus ℓ and w_ℓ a free scalar parameter. The gate vector w carried one entry per input locus — here 83,527 learnable scalars — and rather than starting from scratch we initialized it from the foundation model's VES scores, so that training began from an evolution-informed prior on locus importance. Writing s_ℓ for the VES score of locus ℓ, the initial gate logit was the doubled z-score of the log-absolute score,

$$w_\ell^{(0)} = 2\,\frac{\log|s_\ell| - \mathrm{mean}_j \log|s_j|}{\mathrm{sd}_j \log|s_j|},$$

with $|s_\ell|$ floored at 10^{-8} before the logarithm and the standard deviation floored at 10^{-6} for numerical stability. This transform placed $\sigma(w_\ell^{(0)})$ in approximately $[0.02, 0.98]$: loci the felid foundation model found most constrained (high $|s_\ell|$) started with gates near one and effectively passed through, while low-scoring loci started near zero and were initially suppressed. Backpropagation then refined every gate against the geographic labels, letting the task-specific signal adjust the foundation-model prior. Because the gate multiplied the input element-wise, it added only one scalar per locus and was negligible relative to the MLP trunk's parameter count.

After the gate — when present — the input was standardized by a batch-normalization layer over the input features and then passed through the MLP trunk. Each hidden layer applied a linear projection, an exponential linear unit (ELU) activation, and dropout (**?**), and this `[Linear → ELU → Dropout]` block was repeated L times. The depth L (`n_hidden_layers`), the hidden width (`hidden_dim`), and the dropout probability were all set by hyperparameter search, with code defaults of $L = 2$, width 256, and dropout 0.2; dropout was applied after every hidden layer by default, or after a single designated layer when the optional `dropout_after_layer` index was supplied. The trunk output fed two independent linear heads. A coordinate head, `Linear(hidden_dim, 2)`, emitted standardized $(\mathrm{lat}_z, \mathrm{lon}_z)$ predictions that were denormalized to decimal degrees at evaluation, and a biome classification head, `Linear(hidden_dim, 5)`, emitted one logit per Brazilian biome (Amazon, Atlantic Forest, Caatinga, Cerrado, and Pantanal). The biome head was always instantiated and the forward pass always returned its logits; it was disabled purely at the loss level, by setting its weight `cls_loss_weight` to zero, in which case the cross-entropy term was skipped and only the coordinate head received gradient. The reported learnable-gate model was trained coordinate-only in exactly this way (`cls_loss_weight` = 0), with biome membership recoverable post hoc from the predicted coordinate.

Because the number of loci greatly exceeded the number of individuals, the architecture carried an overparameterization guard adapted from GeoGenIE: whenever the requested hidden width exceeded ten times the number of input features, it was multiplied by 0.8 repeatedly until it fell at or below that ceiling (floored at one), with a warning logged. Since the width was searched only over $[200, 512]$, the full-locus ceiling of $10 \times 83{,}527$ lay far above any width the search could propose, so the guard never fired in practice; it chiefly protects the hard-selection mode, where the input dimension can shrink to a few thousand loci.

How the VES scores entered the model was governed by a single `ves_mode` setting with four values. In *none* mode the raw genotype vector was used unchanged, serving as the no-transfer-learning control. In *weighted* mode every locus was retained but scaled by its absolute VES score, $x_\ell = g_\ell \cdot |s_\ell|$, so that more constrained loci contributed more to the input signal. In *selection* mode only the top-K loci by $|s_\ell|$ were kept — a hard dimensionality reduction from the full ~83k loci down to K, searched over $K \in [2000, 5000]$. And in *learnable* mode all loci were retained behind the sigmoid gate described above, letting the foundation model set the initial locus importances and the task refine them. Weighting and selection were applied per fold, after the within-fold genotype imputation computed from the training individuals alone, so that no held-out information leaked into the transform; in learnable mode, likewise, each fold's model was freshly initialized from the (species-external) VES scores. The default mode in the configuration schema was *selection*.

For the reported full-pipeline model (`ves_mode` = learnable, coordinate-only), the hyperparameter search — best trial 63 of 100 — selected a trunk of four hidden layers of width 507 with dropout 0.218, and this was the configuration realized in the final network.

2.7 | Loss Function and Coordinate Normalization

Because the coordinate head regressed continuous latitude and longitude rather than assigning to discrete classes, we first placed those coordinates on a numerically convenient scale, and we did so independently for every leave-one-out fold. Within each fold the normalization statistics — the two means and two standard deviations, one pair for latitude and one for longitude — were fitted from the training individuals alone, that is from the $N - 1$ animals not held out, so that no information about the test individual could leak into the scaling. We computed these statistics by first collapsing each individual's records to a single mean coordinate pair and then taking the unbiased sample standard deviation (dividing by $n - 1$) across those per-individual coordinates. In the genotype pathway each individual contributes exactly one genome, so this per-individual collapse is a no-op here; it is a generic safeguard inherited from the shared coordinate-statistics routine rather than an operation that changes the genotype-model statistics. To keep a degenerate coordinate distribution from collapsing the denominator, both standard deviations were clamped to a floor of 10^{-6}. Coordinates were then expressed as Z-scores, $z_\phi = (\phi - \mu_\phi)/\sigma_\phi$ for latitude ϕ and analogously z_λ for longitude λ, and the head predicted directly in this centered, unit-variance space. Working in Z-score space keeps the head's outputs near zero and its gradients well conditioned for the optimizer, and because the per-fold statistics were retained, any prediction could be mapped back exactly to decimal degrees.

Although the model predicted in Z-score space, we deliberately defined the training objective in geographic (degree) space, so that the quantity being optimized would coincide with the quantity we ultimately reported. Inside the loss we therefore first denormalized the predicted coordinates back to decimal degrees, $\phi = z_\phi\,\sigma_\phi + \mu_\phi$ and likewise for longitude, and then measured the mean great-circle, or haversine, distance between the denormalized predictions and the degree-space ground truth. The haversine computation is fully differentiable, which is what makes it usable as a training loss, and we implemented it with explicit numerical guards: inputs were promoted to `float32`; degrees were converted to radians through the factor $\pi/180$; the intermediate term $a = \sin^2(\Delta\phi/2) + \cos\phi_1 \cos\phi_2 \sin^2(\Delta\lambda/2)$ was clamped to $[10^{-7},\ 1 - 10^{-7}]$ to keep the subsequent inverse sine well defined under floating-point rounding; and the central angle was obtained as $c = 2\arcsin(\sqrt{a})$ and scaled by an Earth radius of $R = 6371.0$ km to yield a distance in kilometers. Taking the square root before the arcsine — that is, $\arcsin(\sqrt{a})$ rather than $\arcsin(a)$ — is essential, since omitting it systematically overestimates distances. Optimizing a true great-circle distance rather than a Euclidean error in Z-score space matters here because the metric of longitude depends on latitude through a cos(latitude) factor: one degree of longitude spans roughly 111 km at the equator but only about 100 km at latitude $-25°$, a distortion that a Euclidean loss in normalized space simply cannot represent.

The per-batch coordinate loss was this mean haversine distance in kilometers divided by 1000, that is expressed in megameters (1 Mm = 1000 km). We adopted the rescaling purely for gradient conditioning: with assignment errors on the order of a hundred kilometers, the unscaled loss would sit in the hundreds, whereas dividing by 1000 places it in roughly the 0.2–0.8 range. This brings the coordinate loss into the same magnitude as the cross-entropy classification term, so that when both heads are active neither one dominates the gradient.

The full objective combined the two heads as a weighted sum, $\mathcal{L} = w_{\text{coord}}\,\mathcal{L}_{\text{coord}} + w_{\text{cls}}\,\mathcal{L}_{\text{cls}}$, where $\mathcal{L}_{\text{coord}}$ is the megameter-scaled haversine loss described above and $\mathcal{L}_{\text{cls}}$ is an optional cross-entropy over the five biome classes (Amazon, Atlantic Forest, Caatinga, Cerrado, and Pantanal). The biome term was evaluated only when the classification head was present and its weight was non-zero; whenever `cls_loss_weight` was set to 0 it contributed exactly zero and the biome head received no gradient, routing the entire training signal to the coordinate regressor. Both weights defaulted to 1.0 in the software. Our recommended, coordinate-only configuration set `cls_loss_weight` = 0.0 — biome membership can be recovered after the fact from the predicted coordinates by spatial polygon lookup — while the coordinate weight `coord_loss_weight` was not fixed but left to the hyperparameter search, sampled log-uniformly over $[0.5, 6.0]$; when the biome head was enabled, `cls_loss_weight` was likewise searched log-uniformly over $[0.1, 3.0]$. For the best single trial of the recommended run this procedure settled on `coord_loss_weight` ≈ 0.826. We note that in the coordinate-only configuration (`cls_loss_weight` = 0), `coord_loss_weight` multiplies the sole active loss term and is therefore close to degenerate: Adam-family updates are approximately invariant to a global rescaling of the loss and AdamW's weight decay is decoupled from it, so its searched value should not be over-interpreted.

2.8 | Cross-Validation and Hyperparameter Optimization

In scenarios with very small sample sizes, we will utilize leave-one-out cross-validation.

Because our jaguar cohort comprised only a small number of whole genomes, a single held-out test split would have been too small to yield a stable error estimate, so we evaluated every configuration with leave-one-out cross-validation (LOOCV) instead. Concretely, we held out each of the 55 individuals in turn, trained the genotype MLP on the remaining 54, and predicted the held-out individual's coordinates; the 55 resulting per-individual great-circle (haversine) errors were then aggregated into the reported metrics. To keep information from leaking across the fold boundary, we recomputed every quantity that depends on the training set inside each fold: the latitude/longitude standardization statistics were fit on the 54 training individuals only, and missing genotypes were imputed from allele frequencies estimated on the training individuals alone before being applied to the held-out one. Each fold trained the network on its 54 individuals as a single full batch—the matrix fits trivially in memory—and reseeded the random state deterministically as the base seed (42) plus the fold index, so that each fold's data splits, imputation and weight initialisation are deterministic (up to the residual GPU-kernel nondeterminism noted in Section 2.11). We summarized each run by the *median* haversine distance across the 55 folds, a statistic robust to the small number of far-outlier assignments, and it served as both our primary metric and our optimization target; alongside it we recorded the mean haversine distance and the fraction of individuals assigned within 250, 500, and 1000 km, the last three chosen to align with the population-genetics benchmark of **?**.

We optimized the model's architecture and training hyperparameters with the Optuna framework

[[TODO]] EXTEND: add reference for the Optuna hyperparameter-optimization framework

, treating the median LOOCV haversine distance as a black-box objective to be minimized. Each Optuna trial executed a complete 55-fold LOOCV under a candidate hyperparameter set and returned the resulting median haversine distance; we created the study with `direction="minimize"` and ran it for 100 trials, all 100 of which completed. We did not set the sampler explicitly—`optuna.create_study` was called with only `direction` and `study_name`—so the search relied on Optuna's default sampler. For a single-objective study this default is the Tree-structured Parzen Estimator (`TPESampler`), and the Optuna version is pinned to ≥ 4.9.0 in `pyproject.toml`.

For the recommended learnable-gate model with haversine loss, each trial drew its hyperparameters from the ranges listed in Table **??**. Because the biome-classification head was disabled for this model (`cls_loss_weight` fixed at 0), the classification-loss weight was held out of the search; had it been active, it would have been tuned over $[0.1, 3.0]$ on a log scale. Likewise, the locus-count hyperparameter `ves_top_k` was only introduced into the search under the selection VES mode (`ves_mode="selection"`, range $[2000, 5000]$) and was therefore inactive here, since the learnable-gate mode retains all 83,527 loci. Within every fold the network was trained with the AdamW optimizer under a cosine learning-rate schedule whose warmup spanned the first 10% of the trial's epoch budget. The best configuration found (trial 63) used 4 hidden layers, 507 hidden units, dropout 0.218, a learning rate of 2.92×10^{-3}, weight decay 1.21×10^{-5}, a coordinate-loss weight of 0.826, and 436 training epochs, achieving a median haversine distance of 149.2 km.

Each trial evaluates a distinct hyperparameter configuration, and because the per-fold random state is reseeded deterministically as the base seed plus the fold index, a trial's LOOCV result is governed by its hyperparameters rather than by random restarts (up to the residual GPU-kernel nondeterminism noted in Section 2.11); the top-ranked trials therefore differ in their inductive biases rather than in random-restart noise. To combine these complementary configurations without any additional training, we formed a top-k ensemble from the completed trials: we ranked all trials by their objective (the median haversine) and, for $k = \min(5, \text{completed trials})$, averaged the five best distinct configurations' predicted latitude and longitude for each held-out individual, took the majority-vote biome, averaged the class logits, and recomputed the haversine error against the true coordinates. We adopted the ensemble as the final model only when its median haversine improved on the best single trial. In our learnable-gate run the top-5 ensemble reached a median of 164.8 km, which did not beat the best single trial (149.2 km), so we reported the best single trial as the headline figure.

We caution that this best-single-trial figure is an optimistic point estimate. It is the minimum of 100 LOOCV medians, all scored on the same 55 folds used to report accuracy, so selecting the smallest of these correlated values capitalises on chance; no nested (outer) cross-validation was performed to correct this selection. We therefore treat the more conservative top-5 ensemble as our primary estimate and report the best single trial alongside it, keeping both numbers throughout.

Finally, the same pipeline supports a fixed-hyperparameter mode, selected by setting the number of Optuna trials to zero, in which a single LOOCV run is performed directly with the hyperparameters specified in the configuration file rather than searched. We used this mode for the Locator-style baseline, which reproduces the fixed architecture and training defaults of **?**: 10 hidden layers of 256 units, the Adam optimizer (**?**) (in place of AdamW) with no weight decay, dropout of 0.25 applied after the fifth hidden layer only, a learning rate of 0.001, and 5000 training epochs. This baseline shares the LOOCV protocol and haversine loss with the tuned models but performs no hyperparameter search, isolating the effect of the architecture and optimization defaults themselves. The complementary no-VES baseline, by contrast, used the identical 100-trial Optuna search and search space described above but operated on raw genotypes (VES mode disabled), so that the gap between it and the learnable-gate model isolates the effect of the VES prior from any difference in tuning budget. As discussed in Section 3.4, this gap is an *upper bound* on the foundation-model contribution rather than a clean measurement of it, because the learnable model also adds a per-locus gate layer that is absent from the no-VES baseline, so the gap folds in the VES initialisation together with that added layer.

2.9 | Baseline Models

To place the VES-guided results in their proper context, and to attribute any performance gain to its correct source, we trained two baseline models. Both operated on exactly the same input as the VES-guided experiments — the cached dense genotype matrix of 55 individuals × 83,527 biallelic loci, with allele dosages encoded as 0, 1 or 2 — and both were evaluated under an identical protocol: 55-fold leave-one-out cross-validation (LOOCV) with a fixed base seed of 42 (each fold using seed + fold index), per-fold missing-genotype imputation in which each missing call was replaced by the sum of two Bernoulli draws at the alternate-allele frequency estimated from the training fold, and per-fold coordinate normalization fit on the 54 training individuals only. Within every fold the shared multilayer perceptron — batch normalization on the input, hidden layers with ELU activations and dropout regularization (**?**), and separate coordinate- and biome-prediction heads — was trained on the 54 training individuals as a single full batch and used to predict the held-out one. In neither baseline did the biome classification head enter the objective (`cls_loss_weight = 0`), so training minimized the great-circle (haversine) coordinate loss alone; the biome accuracies we report for these runs (Locator baseline 14.5%, tuned no-VES baseline 18.2%) therefore reflect an untrained classification head and are not meaningful assignment results. Optimization in every fold followed a cosine learning-rate schedule with a linear warm-up over the first 10% of epochs. The two baselines differ principally in whether the hyperparameters were fixed at the original Locator defaults or searched with Optuna, together with the fixed optimizer (Adam without

Hyperparameter	Range	Type / scale
`n_hidden_layers`	$[3, 4]$	integer
`hidden_dim`	$[200, 512]$	integer, log
`dropout`	$[0.03, 0.25]$	float
`learning_rate`	$[1.5 \times 10^{-3}, 1.5 \times 10^{-2}]$	float, log
`weight_decay`	$[1 \times 10^{-5}, 5 \times 10^{-3}]$	float, log
`coord_loss_weight`	$[0.5, 6.0]$	float, log
`max_epochs`	$[200, 800]$	integer

TABLE 7 | Optuna search space for the learnable-gate genotype MLP. Each trial ran a full 55-fold LOOCV; the objective minimized was the median haversine distance across folds. The biome classification head was disabled for this model, so `cls_loss_weight` was fixed at 0 and excluded from the search.

weight decay versus AdamW with decoupled weight decay) and dropout-placement choices noted in Section 3.3; neither drew on any variant-effect signal (`ves_mode` =`none`), which lets them isolate two distinct contributions.

Locator baseline. The first baseline applies Locator's published fixed hyperparameters (**?**) to our shared genotype MLP, with no hyperparameter search (`optuna_n_trials` = 0). It is important to be precise about what this reproduces: the network here is a GeoGenIE-style MLP — input batch normalisation, ELU activations, and a differentiable haversine coordinate loss — rather than a byte-for-byte re-implementation of Locator's own architecture and objective, which use ReLU activations and a Euclidean coordinate loss (Section 1). What we transfer from Locator is its configuration, not its exact model: the depth, width, dropout placement, optimizer, learning rate and epoch budget listed in Table **??** — ten hidden layers of 256 units each, dropout of 0.25 applied after the fifth hidden layer only, the Adam optimizer (**?**) with no weight decay, a peak learning rate of 0.001, and 5,000 training epochs. Because this configuration is not adapted to our high-dimensional, low-sample regime (83,527 loci, 55 individuals), the baseline is deliberately *not* a fair comparison for isolating the VES contribution; it is, rather, an untuned reference point that carries Locator's defaults onto our data. Across the 55 LOOCV folds it achieved a median haversine error of 199.2 km (mean 377.4 km), assigning 56.4%, 67.3% and 92.7% of individuals within 250, 500 and 1,000 km respectively. The 67.3% falling within 500 km overlaps the 65–69% that **?** obtained on this species; because their figure comes from a SCAT/F_{ST}-panel analysis rather than from Locator, we read this agreement as a cross-method sanity check that our setup produces sensible errors, not as validation of fidelity to the Locator implementation.

Tuned no-VES baseline. The second baseline holds the loss function, the LOOCV protocol and the raw-genotype input identical to the VES-guided experiments, but replaces Locator's fixed defaults with the same Optuna hyperparameter search we used for those experiments

[[TODO]] EXTEND: add reference for Optuna

, again with `ves_mode` =`none`. This isolates the contribution of Bayesian architecture search: set against the Locator baseline it quantifies the gain from tuning alone, while set against the VES-guided models it provides an upper bound on the gain attributable to the felid foundation model's locus-importance signal, holding tuning constant — an upper bound because, as noted in Section 3.4, the learnable per-locus gate layer is itself absent from this no-VES baseline, so the difference confounds the VES initialisation with the addition of that layer. The search ran 100 trials, all 100 of which completed; each trial executed a full 55-fold LOOCV and was scored by its median haversine distance, which Optuna minimized using its default single-objective sampler (the Tree-structured Parzen Estimator, `TPESampler`; Optuna ≥ 4.9.0), and optimization within each fold used AdamW (Adam with decoupled weight decay). The per-trial search space is listed in Table **??**; the classification loss weight was fixed at 0 to match the coordinate-only objective, and the VES-specific parameters were left out of the search. The best single trial (trial 67) reached a median haversine of 173.6 km (mean 274.3 km; 63.6%, 89.1% and 96.4% within 250, 500 and 1,000 km) with the configuration shown in the right-hand column of Table **??**: four hidden layers, 511 hidden units, dropout 0.174, learning rate 2.01×10^{-3}, weight decay 4.41×10^{-5}, coordinate loss weight 2.50, and 563 epochs. As a final step the pipeline ensembled the five best trials by averaging their per-individual coordinate predictions, which nudged the median slightly to 173.0 km (mean 279.5 km; 63.6%, 87.3% and 96.4% within 250, 500 and 1,000 km); we retained this ensemble as the baseline result. Relative to the Locator baseline's 199.2 km, this shallower and wider architecture — four layers rather than ten, 511 units rather than 256, and roughly an order of magnitude fewer epochs — cuts median error by ≈ 13%, showing that a substantial part of the improvement over the original tool comes from adapting the architecture to the small-sample, high-dimensional setting, before any transfer-learning signal is introduced.

2.10 | Evaluation Metrics

Every configuration was evaluated under the same leave-one-out cross-validation (LOOCV) protocol: each of the 55 jaguar genomes was held out in turn and predicted from a model retrained on the remaining 54 individuals, so that a single fold produced one geographic prediction and the full sweep produced 55 independent held-out predictions. All of the metrics reported below were computed on this pooled set of 55 predictions.

Our primary metric was the *median haversine distance*, in kilometres, between each individual's predicted and true sampling coordinates, and we reported the mean haversine distance

TABLE 8 | Fixed hyperparameters of the Locator baseline (published defaults of **?**), run with no Optuna search under 55-fold LOOCV.

Hyperparameter	Value
Hidden layers (`n_hidden_layers`)	10
Hidden units (`hidden_dim`)	256
Dropout	0.25 (after layer 5 only)
Optimizer	Adam (`use_adam` =true)
Learning rate	0.001
Weight decay	0.0
Training epochs (`max_epochs`)	5,000
Coordinate loss weight	1.0
Classification loss weight	0.0
Optuna trials	0 (fixed)
Seed	42

TABLE 9 | Optuna search space for the tuned no-VES baseline (identical to the VES-guided experiments; 100 trials, objective = median LOOCV haversine, default sampler), and the selected best single-trial configuration (trial 67). Ranges marked "log" were sampled on a logarithmic scale.

Hyperparameter	Search range	Best (trial 67)
`n_hidden_layers`	$\{3, 4\}$	4
`hidden_dim`	$[200, 512]$ (log)	511
`dropout`	$[0.03, 0.25]$	0.174
`learning_rate`	$[1.5 \times 10^{-3}, 1.5 \times 10^{-2}]$ (log)	2.01×10^{-3}
`weight_decay`	$[1 \times 10^{-5}, 5 \times 10^{-3}]$ (log)	4.41×10^{-5}
`coord_loss_weight`	$[0.5, 6.0]$ (log)	2.50
`max_epochs`	$[200, 800]$	563
`cls_loss_weight`	fixed at 0.0	0.0

alongside it as a complementary, more outlier-sensitive summary. Because the model regresses coordinates in per-fold Z-score space rather than in degrees, the haversine (great-circle) distance was always evaluated on denormalised, degree-space coordinates: each prediction was first mapped back to decimal degrees through $\text{deg} = z \times \sigma + \mu$, using the latitude and longitude mean and standard deviation fitted on that fold's 54 training individuals only, and the distance was then taken against the degree-space ground truth. Those per-fold normalisation statistics were fitted on one coordinate pair per training individual; because each jaguar contributes a single genome, this per-individual collapse has no effect in the genotype pathway and is retained only as a generic safeguard from the shared coordinate-statistics routine. We used the standard haversine formula with an Earth radius of 6371 km, evaluating $d = 2R \arcsin\sqrt{a}$ with the intermediate term a clamped to $[10^{-7}, 1 - 10^{-7}]$ for numerical stability; the cos(lat) term in a accounts for the convergence of meridians away from the equator, a distortion that a Euclidean distance in Z-score space cannot represent. The median haversine distance doubled as the objective of the hyperparameter search: each Optuna trial executed a complete 55-fold LOOCV run and returned its median haversine distance, and the study was configured to minimise this value.

The per-individual predictions — true and predicted coordinates, the haversine error, and the true and predicted biome for each individual — were written to `loocv_predictions.json`, and the aggregate metrics to `loocv_summary.json`. Medians were computed with `torch.median`, which for an even number of values returns the lower of the two central order statistics rather than their average. This is exact for the 55-fold pooled median, an odd count, but it shifts the reported per-biome medians of the even-sized biomes — Amazon ($n = 18$) and Pantanal ($n = 6$) — relative to the conventional average-of-two-central-values definition.

To situate our results against the SNP-panel study of **?**, we also reported the fraction of individuals assigned within fixed distance thresholds of their true origin. Specifically, we computed the percentage of held-out individuals whose haversine error was at most 250 km, 500 km, and 1000 km, using inclusive comparisons and dividing by the 55 folds. The 500 km threshold is the one emphasised by **?**, whose reported within-500 km localisation of roughly 65–69% we adopted as the comparison anchor. These percentages are a deterministic function of the per-individual haversine errors in `loocv_predictions.json`. The per-trial LOOCV summaries record them directly, in the fields `pct_within_250km`, `pct_within_500km`, and `pct_within_1000km`, but the ensemble summary file omits them — keeping only the median and mean haversine distance, the classification metrics, and the per-biome medians — so for the top-5 ensemble the threshold percentages were recomputed directly from the per-individual errors.

We further broke the geographic error down by biome of origin, grouping the per-individual haversine errors by each individual's true biome — Amazon, Atlantic Forest, Caatinga, Cerrado, and Pantanal — and reporting both the median and the mean error within each group. As with the distance thresholds, the per-trial LOOCV summaries record the per-biome mean directly whereas the ensemble summary records only the per-biome median, so the per-biome mean was recomputed from the per-individual errors for the ensemble. This breakdown supports a like-for-like comparison with the per-biome results of **?** and exposes the difference in resolution between small, isolated populations and large, continuous ones.

Finally, because the genotype MLP always emitted five biome logits, multi-class classification metrics over the five biomes were computed for every run; they are interpretable, however, only when the classification head was actually trained, that is when the classification loss weight `cls_loss_weight` was non-zero. The predicted biome for a held-out individual was the arg-max of its five biome logits, and from these we computed the overall accuracy (the fraction of individuals whose predicted biome matched the truth), the per-class F_1 score for each biome (the harmonic mean of precision and recall, with degenerate classes — those having no true or predicted members, or zero precision-plus-recall — assigned an F_1 of zero), and the macro-averaged F_1 (the unweighted mean of the five per-class scores). In our recommended learnable-gate configuration the classification head was disabled (`cls_loss_weight` = 0.0), so that the coordinate regression head received the entire gradient signal; the classification metrics still emitted for that run (accuracy 0.182, macro-F_1 0.171 in `loocv_summary.json`) therefore reflect an untrained head and are not interpretable as biome-assignment performance. The configuration for that run notes that biome membership could instead be recovered post hoc from the predicted coordinates via spatial polygon lookup, but

[[TODO]] EXTEND: no post-hoc polygon-lookup code path or recovered-biome field is present in the run outputs, so this procedure and any biome-assignment accuracy derived from it could not be confirmed from the sources.

2.11 | Implementation and Reproducibility

We implemented the entire pipeline as a single installable Python package, `jaguar-geo-assign`, targeting Python ≥3.11, <3.12 and built with the `uv_build` backend (≥0.9.28, <0.10.0). The environment is resolved and locked by the `uv` package manager against a committed `uv.lock`, so that a fresh checkout reconstructs the exact dependency graph rather than re-resolving it. A single console script, `jaguar-geo-assign`, is mapped to the entry point `jaguar_geo_assign.cli:main` and exposes every stage of the pipeline through argument subparsers. We enforced code style and import ordering with `ruff` (target Python 3.11, line length 100), and ran the test suite with `pytest`, whose `integration` marker gates the live NCBI and HuggingFace tests so that they remain off by default (`addopts = "-q -m 'not integration'"`). Table **??** lists the core runtime libraries and the lower bounds pinned for each in `pyproject.toml`; the numerical and modelling backbone is `torch`, with `transformers` loading the DNABERT-2 model and tokenizer (**?**) and `accelerate` handling distributed training and mixed precision.

[[TODO]] EXTEND: add software citations for Python, PyTorch, HuggingFace Transformers, Accelerate, scikit-learn, Optuna, and uv

We enforced reproducibility at the configuration boundary rather than leaving it to convention. Every configuration object is a frozen, immutable dataclass (`@dataclass(frozen=True)`), so that once a TOML file has been loaded and validated it cannot be mutated by any downstream stage. Loading is itself a contract-enforcement step that rejects a configuration *before* any pipeline work begins: the loader verifies that all required sections are present, that the windowing arithmetic is internally consistent—for example, the locus-block size and the tokenizer's maximum position embeddings must each be at least as large as the context window—and that the approved constants have not drifted. To keep those constants authoritative, we placed them in a single module, `pipeline_contract.py`, from which the loaders import them rather than hard-coding values inline. Boolean fields are checked with a strict identity test (`type(value) is not bool`) rather than `isinstance`, so that a TOML integer `1` is never silently accepted where a boolean `true` is required. The foundation corpus is locked down in the same spirit: the six approved felid assemblies form a closed registry, and any configuration that references an unknown, duplicated, or mislabelled identifier is rejected, so that adding a species requires a deliberate change to code and tests rather than an edit to a config file.

The DNABERT-2 dependency is pinned to an exact, immutable revision for both the tokenizer and the model weights. The loader requires the identifier to be `zhihan1996/DNABERT-2-117M` and the revision to be the Git commit `7bce263b15377fc15361f52cfab88f8b586abda0`, and it checks that the continued-pretraining configuration pins its `model_revision` to the same commit, so that warm-starts remain reproducible. Because DNABERT-2 ships a custom tokenizer, `trust_remote_code` must be `true` by contract; pinning the revision bounds exactly which remote code that flag is permitted to execute.

All external assets are acquired through a single checksum-aware transfer primitive, `download_with_retry`. Each download descriptor carries an expected digest and algorithm, and once the full file has been written to a `.part` path its streaming digest is recomputed and compared against the expected value. We treat a checksum mismatch as data corruption rather than a transient fault, so it is not retried through the back-off loop; genuine network failures, by contrast, are retried up to three times with exponential back-off. Interrupted transfers resume from the partial file via an HTTP `Range` header, and a file already present with a matching checksum is skipped entirely, at zero network I/O. The felid reference manifest threads each assembly's expected digest into every download descriptor and asserts that no asset is ever emitted with an empty checksum, so that a future edit cannot silently drop the integrity guard. Within the pinned registry, the five RefSeq assemblies—*Felis catus*, *Panthera tigris*, *Panthera pardus*, *Puma concolor*, and *Panthera leo*—are verified by the MD5 digests taken from each assembly's `md5checksums.txt`, while the DNA Zoo jaguar reference (`DNAZOO_Panthera_onca_HiC`) is verified by a pinned SHA-256 digest (`3b3811dff68a704075cfdceb5fb3f1f4a869d6deda6dced6dd159b6498919229`), guarded by an expected file size of 745,951,926 bytes checked in a HEAD pre-flight, and backed by a HuggingFace mirror that is tried automatically

TABLE 10 | Core runtime dependencies and their version constraints as pinned in `pyproject.toml`.

Package	Version constraint	Role
torch	≥2.11.0	Deep-learning framework
transformers	≥4.45, <5	DNABERT-2 model and tokenizer loading
accelerate	≥1.13.0	Distributed training, mixed precision
pyarrow	≥16, <20	Parquet corpus I/O
scikit-learn	≥1.8.0	Cross-validation **[[TODO: EXTEND: confirm scikit-learn role; README lists StratifiedGroupKFold cross-validation, but no module under src/ imports scikit-learn and the genotype fine-tuning uses a bespoke LOOCV loop]]**
tensorboard	≥2.20.0	Training-metrics logging
beartype + jaxtyping	≥0.22.9 / ≥0.3.9	Runtime type and shape checking
optuna	≥4.9.0	Hyperparameter optimization
einops	≥0.8.2	**[[TODO: EXTEND: confirm einops role]]**

should the primary source fail its checksum.

Data splitting is deterministic and leakage-safe by construction. We partition the genome into locus blocks (50,000 bp in the approved configuration) and assign each block to a fold by hashing its locus identifier: we compute `SHA256("{split_seed}:{locus_id}")`, take the leading 16 hex characters as a 64-bit bucket, normalise it to a position in $[0, 1)$ by dividing by 16^{16}, and select the fold through a cumulative-weight lookup (with default weights `train` = 0.8 and `validation` = 0.2). Because this assignment happens before windowing, every sliding window inherits the fold of its parent block, so that overlapping windows can never straddle the train/validation boundary; the same assignments are reused for the reference-genome baseline. Running the pipeline again with the same seed reproduces identical splits. We note one limit to this determinism: we fixed the random state only through `accelerate`'s `set_seed` (which seeds the Python, NumPy and PyTorch generators) and did not enable PyTorch's deterministic-algorithms mode or the cuDNN determinism flags. Data splits, per-fold imputation, coordinate normalisation and weight initialisation are therefore exactly reproducible, but residual run-to-run nondeterminism arising from GPU floating-point reductions was not disabled.

Model checkpoints are written atomically so that they survive a crash mid-write. Each checkpoint directory is staged into a temporary sibling directory and swapped into place with a context manager, `atomic_dir_replace`, that first moves any existing target aside as a `.old_` backup, then renames the staged directory via `os.replace`, rolling the backup back if the rename fails; a companion recovery routine restores the most recent `.old_` backup after a crash that interrupts the rename. JSON sidecars are written the same way, through `_save_json_atomically`, using a PID-suffixed temporary file and an atomic rename, so that a sidecar is either fully written or absent, never partial. We coordinate these operations across ranks—through `is_main_process` guards and `wait_for_everyone` barriers—so that the scheme remains safe under distributed data-parallel training.

Finally, the tokenized corpus is exported to Parquet with auditability guarantees baked into the export contract. The loader requires genomic coordinates to be preserved (`preserve_coordinates` must remain enabled) and requires that at least one of the raw window sequences or their immutable hashes be retained. In the approved configuration we drop the raw windows (`preserve_raw_windows` = false) but store a SHA-256 hash of each window sequence (`sequence_hash_algorithm` fixed to `sha256`, with `preserve_sequence_hashes` enabled). Storing a per-window hash lets a reader verify after the fact that a materialised corpus matches its source sequences without having to retain the sequences themselves, closing the reproducibility loop from the raw assembly download through to the tokenized training input.

[[TODO]] EXTEND: the actual training hardware (GPU model and count — i.e. the data-parallel world size W in the $32 \times 2 \times W$ effective batch size) and the wall-clock runtimes for foundation pre-training, VES scoring, and the two 100-trial × 55-fold Optuna searches are not recorded in the sources and should be added here.

3 | Results

3.1 | Overall Geographic Assignment Accuracy

We evaluated the complete pipeline end to end — the felid DNABERT-2 variant effect scores used to initialise a set of learnable per-locus importance gates, which in turn feed a genotype multilayer perceptron (MLP) trained under the differentiable haversine loss — over the full 55-fold leave-one-out cross-validation (LOOCV) protocol. Under this protocol we train on all but one of the 55 jaguar whole genomes and predict the held-out individual, cycling through all 55 folds so that every genome is assigned exactly once, across the five Brazilian biomes represented in the sample: Amazon, Atlantic Forest, Caatinga, Cerrado, and Pantanal. The configuration reported here corresponds to `ves_mode = "learnable"` (`configs/examples/genotype_finetune_learnable_haversine.toml`), in which every one of the 83,527 retained loci is passed to the network and the foundation model's evolutionary signal serves only as an initialisation that backpropagation is then free to refine. We selected hyperparameters by Bayesian search, running 100 completed Optuna trials in which each trial was itself a complete 55-fold LOOCV run scored by its median haversine distance; the final model we report is the top-5 ensemble, formed by averaging the per-individual predicted coordinates of the five best-scoring trials.

Across the 55 held-out genomes, the top-5 ensemble assigned individuals to their true sampling location with a median haversine error of 164.77 km and a mean of 276.00 km. Because the median assignment error is our primary metric, we also report how the per-individual errors are distributed around it: 61.8% of individuals (34/55) fell within 250 km of their true origin, 85.5% (47/55) within 500 km, and 96.4% (53/55) within 1000 km. We disabled the biome classification loss for this run (`cls_loss_weight = 0.0`), so no discrete-class objective was optimised at all; the biome labels reported here are therefore simply the argmax of the model's untrained biome head, taken as a majority vote across the five ensemble trials. Their near-chance overall accuracy of 18.2% (10/55) and macro-averaged F_1 of 0.171 confirm what we would expect — that the geographic signal is carried by the coordinate-regression objective rather than by any discrete class assignment.

The five trials entering the ensemble had individual LOOCV median errors of 149.18, 170.45, 175.34, 180.87, and 181.99 km. The best of them, Optuna trial 63, reached a median haversine of 149.18 km on its own — markedly sharper than the ensemble median — with a mean of 268.96 km, and placed 63.6% (35/55), 83.6% (46/55), and 96.4% (53/55) of individuals within 250, 500, and 1000 km respectively (biome accuracy 18.2%, macro-F_1 0.160). We report both figures deliberately and distinguish between them throughout: the ensemble is the more conservative, less overfit estimate of assignment accuracy, whereas the single-trial value reflects the single best hyperparameter draw. In particular, the single-trial value is the minimum of 100 LOOCV medians, all scored on the same 55 folds; selecting this minimum over correlated trial scores makes it an optimistic point estimate, and because no nested (outer) cross-validation was run to correct for that selection, we adopt the following convention throughout: we report the best single trial for continuity with the hyperparameter-search literature — and it is the value bolded in the summary tables — but treat the top-5 ensemble as our conservative primary estimate. Table **??** sets the two side by side.

Throughout, we report point estimates only. With $n = 55$ individuals — and as few as five (Caatinga) or six (Pantanal) per biome for two of the populations — small differences in median kilometres and all per-biome comparisons carry substantial sampling variance, and we did not attach confidence intervals to any figure.

[[TODO]] EXTEND: a bootstrap confidence interval for the overall median haversine and a paired Wilcoxon signed-rank test on the 55 per-individual VES vs. no-VES errors are not present in the run outputs; both could be computed from `loocv_predictions.json`.

Two features of these numbers deserve emphasis. First, the ensemble median (164.77 km) is worse than that of the best single trial (149.18 km), which is exactly what we would expect: averaging coordinate predictions across trials with different inductive biases regularises the assignment but sacrifices the sharpest individual fit. This trade-off does not, however, carry over to the 500 km threshold, where the ensemble (85.5%) slightly exceeds the single trial (83.6%), suggesting that ensembling suppresses large outlier errors even as it relaxes the median. Second, the two estimates bracket our headline result: framed as a continuous-coordinate regression in the spirit of **?**, the pipeline places a typical held-out jaguar within roughly 150–165 km of its true origin, and assigns the great majority of individuals to within 500 km — the threshold most relevant for comparison against traditional population-genetic assignment for this species

[[TODO]] EXTEND: exact within-500 km assignment rate and fold-improvement reported by **?** for jaguar — present only in the README characterisation, not in the code/JSON sources of truth

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3.2 | Comparison With Published Methods

The most directly comparable published benchmark for jaguar geographic assignment is the study of **?**, who assembled 58 whole genomes spanning all five Brazilian biomes (Amazon, Pantanal, Cerrado, Atlantic Forest, and Caatinga) and used pairwise F_{ST} to select the most population-informative loci, arriving at a full Jag-SNP panel of 459 markers together with a reduced 84-SNP panel. On these markers they carried out discrete biome assignment with discriminant analysis of principal components (DAPC) and continuous coordinate prediction with SCAT (**?**). They reported that roughly 98% of individuals were correctly assigned to their biome of origin, and that 65–69% could be localised within 500 km of their true sampling site, with resolution highest in the small, isolated Caatinga and Atlantic Forest populations and lowest in the Amazon, where extensive gene flow across continuous habitat flattens the allele-frequency gradients on which assignment depends.

We compared our pipeline against these figures under a matched continuous-assignment framing, drawing every number we report from 55-fold leave-one-out cross-validation (LOOCV) on our 55-genome dataset. Because our pipeline yields two candidate models — the single best hyperparameter-optimisation trial and a five-model ensemble that averages the predictions of the top five trials — we state explicitly which protocol underlies each figure. The headline comparison that follows uses the *best single trial*, namely trial 63, the trial with the lowest LOOCV median among all 100 Optuna trials; for completeness, the five-model ensemble reached a median of 165 km, higher (worse) than the best single trial. That best-trial model applied learnable per-locus Variant Effect Scoring (VES) gates over all 83,527 candidate loci (`ves_mode = "learnable"`) with the biome classification head disabled (`cls_loss_weight = 0`), so that geographic coordinates were the sole training target.

On the continuous-assignment metric that both studies report, our best-trial model localised 83.6% of individuals within 500 km — above the 65–69% range of **?** — while attaining a median error of 149 km (mean 269 km). These are, however, cross-study comparisons drawn on different sample sets (55 versus 58 genomes — our 55 modelled individuals are those that remained after joining the 57-sample deposited VCF to the coordinate metadata, whereas **?** state 58 whole genomes) and under different feature regimes (83,527 VES-gated loci versus a 459- or 84-locus F_{ST} panel), and we read them accordingly: as evidence that the pipeline is competitive with an established, purpose-built jaguar panel rather than as a controlled head-to-head test.

The per-biome breakdown in Table **??** shows that our median error was lower than, or comparable to, the per-biome *mean* errors reported by **?** in every biome; because this sets our medians against their means, the comparison is indicative

TABLE 11 | Overall geographic assignment accuracy of the full pipeline (learnable VES gates, haversine loss) over 55-fold LOOCV. The top-5 ensemble averages predicted coordinates from the five best Optuna trials; the best single trial is Optuna trial 63.

Metric	Top-5 ensemble	Best single trial (63)
Median haversine (km)	164.8	149.2
Mean haversine (km)	276.0	269.0
Within 250 km	61.8%	63.6%
Within 500 km	85.5%	83.6%
Within 1000 km	96.4%	96.4%
Biome accuracy	18.2%	18.2%
Macro-F_1	0.171	0.160

TABLE 12 | Overall LOOCV comparison of our best single trial against the published jaguar benchmark. "n/r" denotes metrics not reported in the source: **?** report the within-500 km rate (65–69%) but not an aggregate median or the 250/1000 km thresholds. Our biome-head accuracy is omitted ("n/c", not comparable) because the best-trial model was configured for coordinate regression with the biome head untrained.

Metric	This work (best trial)	?
Individuals / LOOCV folds	55	58
Continuous-coordinate method	VES-gated genotype MLP	SCAT
Median haversine error (km)	149	n/r
Mean haversine error (km)	269	n/r
Within 250 km	63.6%	n/r
Within 500 km	83.6%	65–69%
Within 1000 km	96.4%	n/r
Biome assignment	n/c	~98% (DAPC)

rather than strictly like-for-like. The two largest apparent gains fell in the Amazon (402 versus 708 km) and the Cerrado (195 versus 493 km) — precisely the biomes in which the panel resolves origin worst, because high gene flow across vast continuous habitat weakens the spatial allele-frequency gradients on which such assignment relies. In the small, genetically isolated Caatinga (86 versus 80 km) and Atlantic Forest (122 versus 125 km) populations our medians essentially matched the published values, consistent with both methods already resolving these well-differentiated populations close to the limit set by within-population geographic spread.

One caveat bounds the interpretation of Table **??**. We did not attempt a like-for-like comparison against the paper's ~98% biome-assignment accuracy for the headline model, because that figure is a DAPC discrete-classification result whereas our best-trial model was optimised purely for coordinate regression with the biome head disabled; its raw biome-head accuracy of 18%, close to the 20% five-class chance level, is therefore not a meaningful counterpart, and reflects the disabled head in the coordinate-only model rather than a limit of the approach. A separate VES-guided configuration that did train the biome head (the *weighted*-VES multitask run described in Section 3.4) reached a biome accuracy of 87.3% (macro-F_1 0.88) at comparable localisation error, indicating that high biome-assignment accuracy is attainable within the same pipeline when the classification head is active. Biome membership could also in principle be recovered post hoc from the predicted coordinates through a spatial polygon lookup, since those coordinate predictions already sit at biome-relevant resolution, but we did not evaluate that recovery accuracy in the present run outputs.

3.3 | Contribution of Bayesian Hyperparameter Optimization

Having established the VES-guided pipeline, we next asked how much of its accuracy stems from hyperparameter optimization alone, independent of the felid foundation model's variant-effect signal. Locator (**?**) ships with a fixed architecture and training recipe calibrated for large-population datasets, and those defaults are typically carried over unchanged into downstream studies. We reasoned that on the small, high-dimensional jaguar cohort—55 individuals genotyped at 83,527 biallelic loci across five Brazilian biomes—a configuration tuned to the data itself could recover a substantial share of the attainable accuracy before any transfer-learning signal was added. To isolate this effect, we constructed two configurations that shared everything but the hyperparameters: the same genotype matrix, the same leave-one-out cross-validation (LOOCV) protocol over all 55 folds, the same haversine coordinate-regression loss, raw genotypes with no variant-effect scoring (`ves_mode="none"`), and the same random seed (42). They differed only in whether the model hyperparameters were held fixed at Locator's defaults or searched.

The first configuration reproduced Locator's published defaults (**?**): ten hidden layers of 256 units, dropout of 0.25 applied after the fifth hidden layer only, the Adam optimizer (**?**) with no weight decay, a learning rate of 10^{-3}, and 5,000 training epochs, run once with hyperparameter search disabled. This baseline achieved a median haversine error of 199.2 km, assigning 56.4% of held-out individuals within 250 km, 67.3% within 500 km, and 92.7% within 1,000 km. The 67.3% within-500 km figure falls inside the 65–69% range that **?** reported for this species; since their figure comes from a SCAT/F_{ST}-panel

TABLE 13 | Per-biome haversine error (km) for our best single trial (medians) versus the per-biome *mean* errors reported by **?** for their 459-SNP panel. The reference column reports means, whereas ours are medians, so the comparison is indicative rather than strictly like-for-like. Per-biome sample sizes n (summing to 55) were derived from the per-biome accuracy denominators in the run output (Amazon 5/18, Cerrado 3/13, Pantanal 1/6, Caatinga 1/5); the Caatinga ($n = 5$) and Pantanal ($n = 6$) values are corroborated in the project documentation, and Atlantic Forest is the residual to 55.

Biome	n	This work (best trial, median)	**?** (mean, 459-SNP)
Amazon	18	402	708
Cerrado	13	195	493
Atlantic Forest	13	122	125
Pantanal	6	119	196
Caatinga	5	86	80

analysis rather than from Locator, we read this agreement as a cross-method sanity check that the baseline produces sensible errors, and use it as an untuned reference point against which to measure the effect of tuning rather than as confirmation of fidelity to the Locator implementation.

The second configuration replaced these fixed defaults with a 100-trial Optuna Bayesian search

[[TODO]] EXTEND: add reference for Optuna (Akiba et al., 2019)

, holding the loss, the LOOCV protocol, and the genotype input identical to the baseline. Each trial ran a complete 55-fold LOOCV and was scored by its median haversine distance, which the search minimized directly using Optuna's default Tree-structured Parzen Estimator sampler. The search space spanned the number of hidden layers (3 or 4), the hidden width (200–512 units, log scale), dropout (0.03–0.25), the learning rate (1.5×10^{-3} to 1.5×10^{-2}, log scale), weight decay (10^{-5} to 5×10^{-3}, log scale), the coordinate-loss weight (0.5–6.0, log scale), and the number of training epochs (200–800); tuned trials optimized with AdamW under a cosine schedule with 10% warmup. Once the 100 trials had completed, we assembled the final model by averaging the coordinate predictions of the five best trials into an ensemble.

Bayesian optimization produced a marked improvement over the fixed-default baseline (Table **??**). Strictly, the tuned configuration differs from the Locator baseline in two respects beyond the searched hyperparameters: it optimises with AdamW (Adam with decoupled weight decay) rather than plain Adam, and it applies dropout after every hidden layer rather than after the fifth layer only. Neither of these is part of the Optuna search space, so the comparison in Table **??** bundles the optimizer and dropout-placement changes together with the searched hyperparameters. The best single trial (trial 67 of 100) lowered the median haversine error from 199.2 km to 173.6 km, a 12.8% reduction, and raised the fraction of individuals placed within 500 km from 67.3% to 89.1%—a gain of 21.8 percentage points. The top-5 ensemble that we retained as the deliverable model gave a median of 173.0 km and 87.3% within 500 km. The search converged on an architecture strikingly different from Locator's defaults: 4 hidden layers rather than 10, 511 hidden units rather than 256, and 563 training epochs rather than 5,000, together with a dropout of 0.17, a learning rate of 2.0×10^{-3}, a weight decay of 4.4×10^{-5}, and a coordinate-loss weight of 2.50. In other words, the optimizer favored a shallower, wider network trained for roughly an order of magnitude fewer epochs than the Locator defaults prescribe—a configuration far better matched to the high-dimensional, low-sample regime of this dataset, where the deep, narrow default is prone to overfitting. This result indicates that a large share of the accuracy attainable on such cohorts can be unlocked by hyperparameter search alone, before any cross-species transfer signal is introduced, and that this gain is directly transferable to existing Locator-based conservation-genomics workflows without any methodological change.

3.4 | Contribution of Variant Effect Scoring

To isolate the contribution of Variant Effect Scoring (VES) from the improvement attributable to hyperparameter optimization alone, we trained a matched no-VES baseline that was identical to the full pipeline in every respect but one — the treatment of loci. Both configurations operated on the same genotype matrix of 55 individuals and 83,527 candidate SNPs, drew from the same MLP architecture search space, minimized the same differentiable haversine coordinate loss (with the biome-classification head disabled, `cls_loss_weight` = 0), followed the same 55-fold leave-one-out cross-validation (LOOCV) protocol, and were tuned under the same 100-trial Optuna budget with a fixed seed of 42. The two runs differed only in the configuration flag `ves_mode`. The baseline set `ves_mode="none"` and consumed the raw genotype vector directly, whereas the VES-learnable model set `ves_mode="learnable"` and applied a per-locus gate of the form $x = \text{genotypes} \times \sigma(g)$, whose logits g were initialized from the felid-pretrained DNABERT-2 (**?**) VES scores and then refined by backpropagation during training. Because everything else was held constant, the gap in performance between the two is an *upper bound* on the VES-specific contribution rather than a clean measurement of it: the `ves_mode="none"` baseline has no gate at all, whereas the learnable model adds an 83,527-parameter per-locus sigmoid gate, so the difference confounds the evolutionary content of the VES initialisation with the mere addition of a learnable per-locus scaling layer. We did not run a gate-with-uninformative-initialisation control (for example a random- or constant-initialised gate of the same size), and we did not analyse the trained gates to confirm that the VES structure survives optimisation; the gap should therefore be read as bounding, not isolating, the transfer-learning signal.

On a like-for-like comparison of the single best Optuna trial from each search, the no-VES baseline (trial 67) reached a median LOOCV haversine error of 173.6 km, while the VES-learnable model (trial 63) reached 149.2 km, a reduction

TABLE 14 | Effect of Bayesian hyperparameter optimization on geographic assignment, holding the loss, LOOCV protocol, and raw-genotype input constant. The Optuna-tuned column reports the best single trial; the top-5 ensemble deliverable gave 173.0 km and 87.3% within 500 km.

Metric	Locator baseline	Optuna-tuned	Improvement
Median haversine (km)	199.2	173.6	−12.8%
Within 500 km	67.3%	89.1%	+21.8 pp

of 24.5 km or 14.1%. The two winning trials happened to share the same trunk depth of four hidden layers but differed elsewhere: the no-VES trial used a hidden width of 511 units trained for 563 epochs, whereas the VES-learnable trial used 507 units and 436 epochs. Since selecting a single best trial can reward a fortunate initialization, we also compared five-trial ensembles that averaged the predictions of the five best trials from each search (`ensemble_k` = 5). Under this more conservative view the improvement persisted but shrank considerably: the median haversine error fell from 173.0 km to 164.8 km, a 4.7% reduction, and the mean haversine error was essentially unchanged, moving from 279.5 km to 276.0 km, a 1.3% reduction. The VES prior therefore helped consistently, but most of the headline gain was concentrated in the best-performing trial rather than realized uniformly across the top of the trial distribution.

This median improvement, moreover, did not extend to the within-500 km threshold, on which the tuned no-VES model was in fact the better performer: 89.1% (best trial) and 87.3% (ensemble) fell within 500 km, against 83.6% and 85.5% for the VES model. The VES contribution is thus specific to the median error and does not hold on the within-500 km rate — the very threshold shared with **?** — so the improvement should be read as metric-dependent rather than uniform.

We also note that the learnable coordinate-only configuration reported here was not the numerically best VES run in our experiments. A *weighted*-VES variant that additionally trained the biome-classification head (`ves_mode="weighted"`, `cls_loss_weight` ≈ 1.48, i.e. a genuine multitask objective) reached a slightly lower best-trial median of 146.3 km and a top-5 ensemble median of 150.3 km, and, because its biome head was trained, it also attained a meaningful biome accuracy of 87.3% (macro-F_1 0.88). We nonetheless headline the learnable coordinate-only model because it holds the loss and objective identical to the no-VES and Locator baselines, so that the 199 → 174 → 149 km chain isolates the effect of the VES prior under a fixed coordinate-only protocol rather than confounding it with the addition of a multitask classification term. The weighted variant is disclosed here to document that a VES-guided run with a trained biome head reaches comparable localisation accuracy while also assigning biome at ~87%, and it is this run that underlies the biome-accuracy figure cited in Section 3.2.

To see where in the range the transfer-learning signal acted, we broke the best-trial error down by true biome (Table **??**). The largest relative improvement fell on Caatinga (n = 5), the smallest and most data-poor population, where the median error dropped from 94.5 km to 85.6 km (9.4%). Atlantic Forest (n = 13) improved from 131.9 km to 121.9 km (7.6%) and Amazon (n = 18) from 421.4 km to 401.6 km (4.7%). In the two remaining biomes the VES model was marginally worse: Pantanal (n = 6) rose from 117.0 km to 119.5 km (2.1% worse) and Cerrado (n = 13) from 187.2 km to 195.4 km (4.4% worse). Among the two smallest populations, then, only Caatinga showed a clear gain while Pantanal was essentially unchanged — a more nuanced pattern than a uniform small-population benefit.

This per-biome picture proved sensitive, however, to how each configuration was summarized. Under the five-trial ensembles the same breakdown (again reported as per-biome medians) reordered substantially: VES improved Atlantic Forest (126.9 km to 115.2 km), Cerrado (188.3 km to 164.1 km) and Pantanal (147.8 km to 141.0 km), yet regressed sharply in Amazon (327.3 km to 397.7 km) and slightly in Caatinga (106.4 km to 108.8 km). The disagreement between the best-trial and ensemble views tells us that, at 55 individuals with as few as five per biome, per-biome error estimates carry high sampling variance and should be read as indicative rather than definitive. Taken together, the results show that VES delivered a modest but genuine reduction in overall median assignment error, and that under the best-trial analysis the data-scarcest population benefited most — consistent with the premise that a label-free, cross-species constraint prior is most valuable precisely where within-sample allele-frequency estimates are least reliable.

3.5 | Per-Biome Assignment Accuracy

The aggregate median conceals a strong biome-to-biome structure, so we broke the leave-one-out errors down by the true biome of each held-out individual. The five Brazilian biomes differ sharply in spatial extent, connectivity, and the number of reference genomes available, and the held-out set reflected that imbalance: it contained 18 Amazon, 13 Atlantic Forest, 13 Cerrado, 6 Pantanal, and 5 Caatinga individuals, 55 in total. This breakdown also let us compare our results directly, biome by biome, against **?**, who reported the assignment error of their 459-SNP jaguar panel separately for each biome.

Two aspects of what we report bear on reproducibility. First, the figures in the "Ensemble" column are the per-biome medians written to the `per_biome_haversine` field of the ensemble's LOOCV summary; they are computed on the coordinate predictions obtained by averaging the five best Optuna trials (`ensemble_k` = 5), and they are medians—not means—of the per-individual great-circle errors within each biome. Second, because the ensemble and the single best trial disagree at the biome level, we also report the per-biome medians of the best individual trial (trial 63, the trial that minimises the overall median at 149.2 km), taken from its `per_biome_haversine_median` field. The two sets differ substantially for some biomes, so conflating them would misrepresent the result. The reference column reproduces the per-biome figures of **?** for their 459-SNP panel; we note that

TABLE 15 | Contribution of VES transfer learning, measured as the change in LOOCV haversine error when moving from the raw-genotype baseline (`ves_mode="none"`) to VES-guided learnable locus gates (`ves_mode="learnable"`). Both were tuned with the same 100-trial Optuna budget, search space, loss, and 55-fold LOOCV protocol.

Comparison	No-VES	VES-learnable	Improvement
Best single trial, median (km)	173.6	149.2	14.1%
Best single trial, mean (km)	274.3	269.0	2.0%
Five-trial ensemble, median (km)	173.0	164.8	4.7%
Five-trial ensemble, mean (km)	279.5	276.0	1.3%

TABLE 16 | Per-biome median haversine error (km) for the best single trial of each configuration, with per-biome sample sizes. Improvement is the relative reduction in error from the no-VES baseline to the VES-learnable model; negative values indicate the baseline was better.

Biome (n)	No-VES	VES-learnable	Improvement
Caatinga ($n = 5$)	94.5	85.6	9.4%
Pantanal ($n = 6$)	117.0	119.5	−2.1%
Atlantic Forest ($n = 13$)	131.9	121.9	7.6%
Cerrado ($n = 13$)	187.2	195.4	−4.4%
Amazon ($n = 18$)	421.4	401.6	4.7%

those are reported by the authors as *mean* distances, whereas ours are medians, so the comparison is indicative rather than strictly like-for-like.

The per-biome pattern was consistent across both aggregations and mirrored the behaviour reported for the traditional panel. The Amazon, the most spatially extensive of the five biomes, was by far the hardest to localise (ensemble median 397.7 km; best-trial median 401.6 km), whereas the small, isolated, and genetically impoverished Atlantic Forest and Caatinga populations were localised most precisely (ensemble medians of 115.2 and 108.8 km). This ordering matches the observation of **?** that resolution was lowest in continuous habitats such as the Amazon—attributed to high gene flow and correlated allele frequencies—and highest in isolated, low-diversity populations.

The two biomes the panel localised worst, the Amazon and the Cerrado, were precisely where our pipeline gained the most. Both aggregations placed the Amazon median roughly 300 km below the panel mean (ensemble 397.7 versus 707.67; best trial 401.6) and the Cerrado median roughly 300 km below it (ensemble 164.1 versus 492.62; best trial 195.4). In the remaining three biomes the two sets of figures were of the same order: the Atlantic Forest medians (115.2 and 121.9) sat close to the panel mean of 125.36; the Pantanal medians (141.0 and 119.5) fell below the panel mean of 195.56; and in the Caatinga—the biome the panel localised best—our errors were comparable to the panel mean of 80.45 km, with the best-trial median (85.6 km) essentially matching it and the ensemble median (108.8 km) running somewhat higher. We stress that these comparisons mix our medians with the paper's means, so the results in the Atlantic Forest, Pantanal, and Caatinga should be read as "comparable" rather than as strict wins; in the Caatinga, in fact, the ensemble median slightly exceeds the panel mean.

The ensemble and the best trial diverged most in the Cerrado (164.1 km ensemble versus 195.4 km best trial) and, in the opposite direction, in the Caatinga and Pantanal, where the best trial was sharper (85.6 and 119.5 km) than the ensemble (108.8 and 141.0 km). This within-pipeline variability is a direct consequence of the very small per-biome sample sizes—$n = 5$ for Caatinga and $n = 6$ for Pantanal—which make the per-biome medians sensitive to individual folds. It also explains why the trial with the lowest *overall* median (trial 63, at 149.2 km) is not uniformly the best trial within every biome, and why its overall median is lower than that of the ensemble (164.8 km) even though ensembling improves the per-biome medians in the Amazon, Atlantic Forest, and Cerrado.

We summarize the geographic-assignment performance of every configuration in Table **??**, reporting the two metrics used throughout the jaguar assignment literature: the median great-circle (haversine) distance between predicted and true sampling coordinates across the 55 leave-one-out cross-validation (LOOCV) folds, and the percentage of individuals assigned within 500 km of their true origin, the primary accuracy threshold reported by **?**. Because median haversine distance was the objective minimized during Bayesian optimization, for the two Optuna-tuned configurations we report the single best trial—the minimum LOOCV median over the 100 trials—as the headline value, and give the corresponding five-trial ensemble equivalents in the table note.

Taken together, these results show that our two contributions acted on complementary axes of performance. The best single VES-learnable trial achieved the lowest median haversine distance overall, at 149.2 km, whereas the largest single gain in within-500 km accuracy came from Bayesian optimization of the Locator architecture alone, which lifted the fraction of individuals assigned within 500 km from 67.3% to 89.1%.

[[TODO]] EXTEND: error-distribution map not generated in repo outputs

TABLE 17 | Per-biome geographic assignment error under leave-one-out cross-validation. "Ensemble" and "Best trial (63)" are the median haversine distance (km) per true biome for the full VES-guided pipeline (five-best-trial ensemble and best single Optuna trial, respectively); the two differ because they aggregate different sets of predictions. The reference column gives the per-biome *mean* distances reported by **?** for their 459-SNP panel. All distances in km.

Biome	*n*	Ensemble (median)	Best trial 63 (median)	**?** (mean, 459-SNP)
Amazon	18	397.7	401.6	707.67
Atlantic Forest	13	115.2	121.9	125.36
Cerrado	13	164.1	195.4	492.62
Pantanal	6	141.0	119.5	195.56
Caatinga	5	108.8	85.6	80.45
Overall (55 folds)	55	164.8	149.2	—

TABLE 18 | Geographic-assignment performance across configurations, computed over 55 LOOCV folds on the jaguar genotype matrix. Values for the two Optuna-tuned configurations are the single best trial (minimum LOOCV median across 100 trials); their top-five-trial ensemble equivalents are given in the note. Traditional-method figures are the reference values reported for SCAT by **?**.

Configuration	Median haversine (km)	Within 500 km (%)
Traditional methods (SCAT; **?**)	~350–400	65–69
Locator baseline (defaults; **?**)	199.2	67.3
+ Bayesian optimization (no-VES tuned)	173.6	89.1
+ VES learnable locus gates	**149.2**	83.6

Note. For the two Optuna-tuned configurations we also evaluated an ensemble that averages the coordinate predictions of the five best trials by LOOCV median (the top-5 ensemble). The VES-learnable ensemble gave 164.8 km / 85.5%, and the no-VES tuned ensemble gave 173.0 km / 87.3%. Relative to the corresponding best single trial, the VES-learnable ensemble raised the median (164.8 vs. 149.2 km) while slightly improving within-500 km accuracy (85.5% vs. 83.6%), whereas the no-VES tuned ensemble lowered the median marginally (173.0 vs. 173.6 km) at a slightly lower within-500 km accuracy (87.3% vs. 89.1%). We computed the ensemble within-500 km percentages from the per-fold predictions, since the ensemble summary outputs report only the median. The Locator baseline was a single fixed-hyperparameter LOOCV run with no Optuna search, so the best-trial-versus-ensemble distinction does not apply to it. The traditional-method SCAT figure of ~350–400 km median is a narrative characterisation of **?** taken from the project documentation rather than a value recomputed from our code/JSON sources of truth, and should be read together with the within-500 km caveat noted in the Overall Geographic Assignment Accuracy results (Section 3.1); it is provided only for orientation.

4 | Discussion

[[TODO]] PH-W (Discussion): interpret the results around four contribution threads — (1) Bayesian optimisation unlocks Locator on small samples; (2) the felid foundation model as a reusable asset; (3) VES as a label-free alternative to F_ST; (4) transfer learning helps most where data is scarcest — plus limitations and future work. Draw on README "Key Contributions"; the proposal "EXPECTED OUTCOMES" (chapter-proposal.tex L157–170) may be used as idea material only, not copied.

Author Contributions

[[TODO]] PH-X (Author Contributions): CRediT roles per author.

Acknowledgements

[[TODO]] PH-Z (Acknowledgements): funders, LBGM-PUCRS, and field collaborators (the proposal acknowledgements block is empty).

Ethics Statement

[[TODO]] Ethics Statement: sampling and collection permits and animal-ethics approvals for the jaguar genomes (permit numbers and issuing authorities; not present in the proposal sources).

Conflict of Interest

[[TODO]] PH-AA (Conflict of Interest): standard declaration.

Data Availability Statement

The felid foundation model was pretrained on six publicly available reference assemblies spanning four genera of the Felidae (Table **??**). Five of the six were retrieved from the NCBI RefSeq FTP archive (`ftp.ncbi.nlm.nih.gov/genomes/all/GCF`), where each is addressed deterministically by its RefSeq assembly accession and assembly name, so that the same file is recovered on every download. The sixth genome, the jaguar (*Panthera onca*) reference, is not carried by RefSeq: we used instead the Hi-C scaffolded assembly `Panthera_onca_HiC` distributed by DNA Zoo, downloaded from `https://dnazoo.s3.wasabisys.com/Panthera_onca/Panthera_onca_HiC.fasta.gz` (a compressed FASTA of 745,951,926 bytes), and we deposited a mirror at the HuggingFace dataset `coding-racoon/jaguar-reference-dnazoo`, pinned to revision `470b6aae8eabeaf7a6bf9c841583be4b271519ca`. Every reference download was integrity-checked against a checksum pinned in the analysis code — MD5 digests for the five RefSeq FASTAs and a SHA-256 digest for the DNA Zoo assembly — and because the species list is itself closed and hard-coded, the composition of the pretraining corpus is fixed and reproducible.

The two jaguar-specific inputs that drive the fine-tuning path are hosted together as a single public, credential-free HuggingFace dataset (`coding-racoon/jaguar-raw-data`). The

TABLE 19 | The six felid reference assemblies used as the foundation pretraining corpus. RefSeq accessions resolve on the NCBI RefSeq FTP archive; the jaguar assembly is the DNA Zoo Hi-C reference.

Species	Common name	Assembly	Accession / source	NCBI taxon ID
Felis catus	Domestic cat	Felis_catus_9.0	GCF_000181335.3 (NCBI RefSeq)	9685
Panthera leo	Lion	P.leo_Ple1_pat1.1	GCF_018350215.1 (NCBI RefSeq)	9689
Panthera tigris	Amur tiger	PanTig1.0	GCF_000464555.1 (NCBI RefSeq)	74533
Panthera onca	Jaguar	Panthera_onca_HiC	DNA Zoo	9690
Puma concolor	Puma	PumCon1.0	GCF_003327715.1 (NCBI RefSeq)	9696
Panthera pardus	Leopard	PanPar1.0	GCF_001857705.1 (NCBI RefSeq)	9691

first is a multi-sample variant call file (VCF), `jaguar.57samples.allChr.snps.hardFilter.bi.maf.ld.masked.hwe.recode.vcf` (147,354,171 bytes, roughly 147 MB), which records 57 jaguar samples genotyped at biallelic SNPs that had already been hard-filtered and cleaned for minor-allele frequency, linkage disequilibrium, and Hardy–Weinberg equilibrium. This VCF was originally published on DataDryad under `doi:10.5061/dryad.4tmpg4fkm` (CC0 license), and the SHA-256 pin recorded in our code was cross-validated against the digest returned by the DataDryad API. The second input is the sample-metadata table `jaguar_location.csv`, which ties each sample to its geographic coordinates and biome label through the columns `sample_id`, `individual_id`, `latitude`, `longitude`, and `biome_population_label`. Both files were downloaded without credentials from the dataset's default (`main`) branch and verified against SHA-256 digests pinned in the code; unlike the DNA Zoo mirror, however, they are not addressed by a fixed commit revision, so their integrity rests solely on those pinned checksums.

The underlying genomic language model, DNABERT-2 (**?**), is publicly available from the HuggingFace model hub (`zhihan1996/DNABERT-2-117M`) and was pinned to revision `7bce263b15377fc15361f52cfab88f8b586abda0` for exact reproducibility. Finally, all of the analysis code — data acquisition with checksum verification, corpus construction and tokenization, foundation pretraining, Variant Effect Scoring, and the genotype-MLP fine-tuning and evaluation pipeline — is available in the accompanying repository

[[TODO]] EXTEND: add public URL/DOI for the analysis code repository (genomic-transformer-transfer-learning); no repository URL is declared in the sources

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[[TODO]] Citation cleanup: reused proposal prose sometimes names an author and year inline (e.g. "Battey and colleagues (2020)") in addition to the \citep citation, which now renders a redundant parenthetical under author–year style. Reconcile these to \citet / \citep as appropriate — a citation-command change only, not a wording change.

Supporting Information

[[TODO]] PH-AB (Supporting Information): optional; list supplementary tables/figures if any.